

Who Gains from Globalisation? State Capacity and the Effective Taxation of Capital and Labour*

Céline Azémar^a, Rodolphe Desbordes^b, and
Markus Eberhardt^c

^aRennes School of Business

^bSKEMA Business School, Paris and Université Côte d'Azur, Nice

^cUniversity of Nottingham and Centre for Economic Policy Research

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Abstract: Does opening to trade and capital flows strengthen or weaken governments' ability to tax? Using *de jure* trade and capital-account liberalisation policies, we show that the effect depends on state capacity. Contrary to recent work, liberalisation does not raise capital taxation in developing countries while depressing it in rich economies. Instead, it weakens the effective capital tax rate where state capacity is low and lifts it in high-capacity states, implying that the gains from globalisation mainly accrue to richer countries. The same mechanism also shapes labour taxation. What looked like a developing-world success is instead a win for high-capacity states.

Keywords: Effective tax rates; globalisation; factor reallocation; interactive fixed effects; development heterogeneity.

JEL codes: F62, H26, H87, O11

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1 Introduction

Does opening up to trade and capital flows result in more tax revenue for the government? Tax-competition models ([Zodrow and Mieszkowski 1986](#), [Wilson 1986](#), [Devereux, Lockwood, and Redoano 2008](#)) predict a race to the bottom as capital becomes mobile in liberalised economies. Recent evidence for developing economies, however, has found the opposite effect: [Bachas et al. \(2023\)](#) show that since the 1990s, effective capital tax rates have *risen* across the developing world, arguing that trade integration draws economic activity into the larger, formal firms where capital is taxed more effectively, thus widening the tax base. Whether a wider base turns into higher revenue, though, depends on something these authors set aside: whether the state has the *ability to collect taxes*. That ability, state capacity, varies enormously across countries, and most of all among developing countries, where the fiscal promise of opening up is supposed to have already become reality.

In this paper, we ask which countries actually end up successfully taxing more when they liberalise, and find that the answer turns on their administrative capacity: in countries where the state already has the administrative reach to collect taxes, opening up lifts the effective taxation of *both* capital and labour, and, where it does not, the capital tax base is eroded. The fiscal gains from globalisation hence accrue to those economies that could already tax, whereas the poorest states, those most in need of additional revenue, capture little of it or even register a net fiscal loss, due to low state capacity. What looks in the aggregate like a developing-world success is instead a win for high-capacity countries, drawing a continuous gradient in the capacity to tax rather than a divide between rich and developing countries.¹ The effect of trade liberalisation on the capital effective rate runs from about -2 percentage points for the bottom tercile of the capacity distribution to nearly $+3$ for the top one. Capital-account liberalisation matches the ‘penalty’ at the bottom and reaches $+2.3$ points at the top. Although somewhat moderated, these patterns are repeated for the labour effective rate.

We reach these results adopting a different treatment to [Bachas et al. \(2023\)](#) and an empirical design matched to the data. Where these authors measure globalisation by *de facto* trade, the volume of imports and exports relative to output, we use *de jure* liberalisation: the statutory removal of trade barriers and, separately, of capital-account restrictions,² the classic tax-competition channel they set aside. Rather than sorting countries into rich and developing, we let each have its own response to liberalisation and relate that response to state capacity, which the public-finance literature takes to be the binding constraint on what and whom a state can tax ([Jensen and Weigel 2025](#)). Our object of interest is the *capacity*

¹The poorest countries and those with the lowest state capacity do not overlap as much as one might think: for the lowest and middle terciles *by income*, only 56% of countries are in the same *capacity* tercile, whereas 43% are one tercile removed (e.g. low-income tercile country with middle-tercile capacity), with 1%, a single country, two terciles away. If we repeat this exercise for the middle and highest income terciles, these figures are 64%, 33%, and 3%; see Appendix Section D.2.

²The statutory reforms we track, using the Measure of Aggregate Trade Restrictions index ([Estefania-Flores et al. 2022](#)) and the [Chinn and Ito \(2006\)](#) index, can be shown to raise the *de facto* trade volumes used in the analysis of [Bachas et al. \(2023\)](#).

gradient: how the effect of liberalisation on factor taxation shifts as we move from low- to high-capacity economies. We proxy state capacity by a non-fiscal index of impartial administration, the rule of law, and territorial reach (O'Reilly and Murphy 2022),³ so that the gradient we estimate cannot be a mechanical reflection of measuring a tax response with a tax-based proxy. Country-specific liberalisation effects come from a difference-in-differences framework that allows common shocks to affect economies differentially (Pesaran 2006, CCE), which also avoids the biases standard pooled estimators were recently found to incur when treatment is staggered and effects are heterogeneous (Callaway and Sant'Anna 2021, Goodman-Bacon 2021, Sun and Abraham 2021, Borusyak et al. 2024).

The positive state capacity gradient holds for both reforms, trade and capital-account liberalisation, and on the capital and labour effective rate, respectively. A country's capital and labour responses move together rather than offsetting each other, so, provided the state has sufficient capacity to tax, liberalisation imposes no particular pressure on taxing the immobile (labour) factor *more* to make up any revenue shortfall from mobile (capital) factor taxation due to a supposed 'race to the bottom'. The underlying origin for this capacity-graded change in capital and labour effective taxation can be found among the global multinational firms, which account for the bulk of international trade and of foreign direct investment. We show that opening either channel pulls in direct investment, in fact trade liberalisation actually dominates, raising the FDI-to-GDP ratio by 45%, whereas capital liberalisation raises it by merely 30%. Capable states tax this common opening (common to high- and low-capacity states) more effectively, rather than attracting a more substantive base to tax; states with low capacity, in contrast, see mobile capital escape and their base erode. Finally, liberalisation does move the capital and labour income shares, in the equal-and-opposite way Stolper and Samuelson (1941) predict, and this share response tracks a country's capital abundance.

The patterns we uncover do not depend on our specific (CCE) estimation approach: they appear in a two-way fixed-effects regression where we interact liberalisation with state capacity, in a simple mean-group estimator, and in alternative interactive-fixed-effects models that estimate the common shocks rather than proxying them by cross-section averages as in our Pesaran (2006) common correlated effects (CCE) implementation. They are also not dependent on our specific definition of liberalisation, or that of state capacity, or the treatment of social security contributions in the analysis of the labour ETR, and are unlikely to be driven by consumption tax reform or pre-liberalisation trends.

Our conclusions refine the insights of Bachas et al. (2023) rather than overturning them. Their mechanism works through formalisation and should be strongest where the informal sector is largest; they identify it in large middle-income economies with event studies and using instrumental variable regression, finding a positive effect there and an insignificant or negative one in rich countries, which they attribute to globalisation (trade liberalisation). We do not dispute the basic facts, that capital tax rates rose in poor countries and fell in rich ones, but we read the underlying cause differently. Much of the developing-world rise is a

³Robustness checks confirm that our findings do not hinge on this particular state-capacity measure.

secular trend shared by liberalising and non-liberalising countries alike, and where liberalisation does bite it is graded by state capacity, so the developing-versus-rich divide is a continuous capacity gradient seen at two points. These authors' optimistic reading holds for the capable middle-income economies (MICs) they study, but breaks for the poorest countries, which their dichotomy places with the MICs on the winning side of the divide. Instead, our state capacity mediator frequently separates the latter from the former. Nor is the fall in rich-country capital taxation a consequence of opening up, since we show that liberalisation *raises* the effective rate wherever capacity is high (and 80% of countries in the top income tercile are also in the top capacity tercile).⁴

We make three contributions. First, a parsimonious specific-factors model with an explicit tax-capacity technology yields clear, testable hypotheses: the effect of liberalisation on each factor's taxation is a sign-restricted gradient in state capacity. Second, we test these hypotheses with an estimator matched to the data: the sample period is dominated by global shocks and the prior literature suggests that country effects are heterogeneous, so we use a common correlated effects mean-group estimator ([Pesaran 2006](#)), which nets out shocks that reach countries unequally and lets each country's response differ; the pooled IV designs of the prior literature can do neither. Identification rests on a factor-spanning assumption, which can be evaluated with the differential cross-sectional-dependence (Δ CSD) test we introduce. Third, we reassess [Bachas et al. \(2023\)](#) and reach different conclusions: the liberalisation effect is graded by capacity, which income proxies only coarsely, so their developing-versus-rich country divide masks a continuous gradient; globalisation is not the main driver of the cross-country rise, and its gains reach the already-capable rather than the poorest; and trade and capital-account liberalisation act through a common channel, each raising the foreign direct investment inflows whose taxation capacity governs.

The remainder of the paper is organised as follows. Section 2 presents the theoretical framework. Section 3 describes the data and sample, before Section 4 lays out the empirical strategy. Our main results are reported in Section 5 and related to the prior literature in Section 6. Section 7 concludes.

2 Theoretical Framework

We use a stylised specific-factors economy with an explicit tax-state capacity technology to work out three testable sign restrictions. The model is deliberately minimal: its sole purpose is to state three primitive monotonicity assumptions and read off from each a clear prediction about how the effect of globalisation on factor taxation varies with the country's developmental state capacity. We do not micro-found the monotonicities; they are the economic content, and the predicted signs are the restrictions they imply.

⁴We set out where the two findings reconcile, and where they part, in Section 6.

2.1 Environment

Country i at date t produces net domestic product $\text{NDP}_{it} = r_{it}K_{it} + w_{it}L_{it}$, the sum of capital and labour income. The capital and labour income shares are $\theta_{K,it} = r_{it}K_{it}/\text{NDP}_{it}$ and $\theta_{L,it} = 1 - \theta_{K,it}$. For each factor the tax system has two parts: a *statutory rate* κ_f set by law, and the *captured share* $\phi_f \in [0, 1]$, the fraction of the factor's income the authority actually reaches and taxes. Suppressing the i, t subscripts, tax revenue τ_f from factor f as a share of NDP and the corresponding effective tax rate ETR_f are

$$\tau_f = \kappa_f \phi_f \theta_f, \quad \text{ETR}_f = \frac{\tau_f}{\theta_f} = \kappa_f \phi_f. \quad (1)$$

The effective tax rate is thus the statutory rate times the share of income captured; revenue additionally accounts for the income share θ_f . Globalisation enters through two treatments that switch on permanently: trade openness $D_{\text{tr},it}$ and capital-account openness $D_{\text{fin},it}$. Let sc_i index the country's *developmental state capacity*, the bundle of capital abundance, administrative reach, and capital mobility that strengthens with development. We measure it empirically using the [O'Reilly and Murphy \(2022\)](#) baseline state-capacity index (impartial administration, rule of law, and territorial reach). Because this index *excludes* any fiscal or tax indicator, the capacity gradient cannot simply mirror a tax-based proxy. Average log income per capita is a coarser proxy. Each of the three assumptions below isolates one channel and ties it to sc_i .

2.2 Assumptions and predictions

Three primitive assumptions, each a standard monotonicity in development, generate the comparative statics. The treatment effect on outcome y_f is $\beta_{f,i} \equiv \partial y_{f,it} / \partial D$, so the cross-derivative $\partial \beta_{f,i} / \partial sc_i$ is the mixed partial $\partial^2 y_f / \partial D \partial sc_i$ whose sign each monotonicity restricts. The monotonicity is what we impose; the sign of the cross-partial is the testable restriction it implies, obtained by the product rule rather than from deeper primitives.

A1 (Factor abundance and Stolper–Samuelson). Let $\omega_i = \ln(K_i/L_i)$ be the (log) capital–labour endowment ratio, increasing in development ($\omega'(sc_i) > 0$), and $\bar{\omega}$ the world average. Trade opening raises the relative reward of the locally-abundant factor, so its effect on the capital share is proportional to relative abundance, $\partial \theta_{K,it} / \partial D_{\text{tr},it} = \psi(\omega_i - \bar{\omega})$ with $\psi > 0$, negative in labour-abundant (low- sc) economies, positive in capital-abundant (high- sc) ones. Hence

$$\frac{\partial^2 \theta_{K,it}}{\partial D_{\text{tr},it} \partial sc_i} = \psi \omega'(sc_i) > 0, \quad (2)$$

with an equal and opposite gradient on $\theta_L = 1 - \theta_K$.

A2 (Tax capacity). Opening up has two effects on the captured share of capital income ϕ_K . It brings newly formal and internationally active capital into the tax base, captured at rate $b(sc_i) > 0$, increasing in capacity ($b' > 0$); and it opens an avoidance and capital-flight margin $\ell(D_{it})$ ($\ell' > 0$), in turn restricted by enforcement $c(sc_i)$ ($c' > 0$). The net response is $\partial\phi_K/\partial D = b(sc_i) - \ell'(D_{it})/c(sc_i)$, *negative* where capacity is low (capture weak, leakage large) and *positive* where it is high, so

$$\frac{\partial^2\phi_{K,it}}{\partial D_{it}\partial sc_i} = b'(sc_i) + \frac{\ell'(D_{it})c'(sc_i)}{c(sc_i)^2} > 0. \quad (3)$$

Since $ETR_K = \kappa_K\phi_K$, the effective rate inherits the sign exactly: $\partial ETR_K/\partial D$ rises from negative (base erosion under weak enforcement) to positive (base capture under strong enforcement) as sc_i grows, the sign-switch that the data display. Revenue $\tau_K = ETR_K\theta_K$ inherits it too, provided the Stolper–Samuelson reallocation of θ_K (A1) does not offset it, which empirically it does not.

A3 (Capture of labour income). Opening up formalises economic activity, drawing labour income into the tax base alongside capital. The effect of opening up on the captured labour share is itself increasing in capacity: $\partial\phi_L/\partial D$ rises with sc_i , exactly as $\partial\phi_K/\partial D$ does in A2 (formalisation net of a leakage restrained by enforcement), so $\partial^2\phi_{L,it}/\partial D_{it}\partial sc_i > 0$. Since $ETR_L = \kappa_L\phi_L$,

$$\frac{\partial^2 ETR_{L,it}}{\partial D_{it}\partial sc_i} = \kappa_L \frac{\partial^2\phi_{L,it}}{\partial D_{it}\partial sc_i} > 0, \quad (4)$$

under *either* form of liberalisation, since both formalise the base. A distinct, classical force acts on the *statutory* rate: as the capital account opens up and capital grows mobile ($m'(D_{fin}) > 0$), the inverse-elasticity rule lowers the revenue-maximising rate on mobile capital and raises it on immobile labour ($\kappa_L^*(m) > 0$). This burden shift is specific to capital-account liberalisation and acts on headline rates, consistent with the fall in the statutory corporate rate we document in Section 6.

Equations (2)–(4) are precisely the gradients $\partial\beta_{f,i}/\partial sc_i$, so the following propositions are their immediate consequence.

Proposition 1 (Factor reallocation) *Under A1, the trade-opening effect on the capital share is increasing in capacity, $\partial\beta_{\theta_K,i}/\partial sc_i > 0$, with an equal and opposite gradient on the labour share.*

Proposition 2 (Capital tax base) *Under A2, the effect of opening up under either form of liberalisation, on capital tax revenue and the capital effective rate is increasing in state capacity, $\partial\beta_{\tau_K,i}/\partial sc_i > 0$ and $\partial\beta_{ETR_K,i}/\partial sc_i > 0$, switching from base erosion at low capacity to revenue formalisation at high capacity.*

Proposition 3 (Capture of labour income) *Under A3, the opening up effect on the labour effective tax rate is increasing in state capacity, $\partial\beta_{ETR_L,i}/\partial sc_i > 0$, under either form of liberalisation.*

Each proposition has the same form, a *capacity gradient*: the country-specific slope $\beta_{f,i}$ is a monotone function of sc_i . The sign of the gradient, not merely of the average effect, is the object our empirical strategy estimates and the discriminating test of the three channels.

The three channels map to the two forms of liberalisation differently, which guides how we read the results. Factor reallocation (A1) is a Stolper–Samuelson prediction specific to *trade* opening and operates on the capital *share*. The capture channels (A2 for capital, A3 for labour) operate through *either* form: both trade and capital-account liberalisation formalise the base, which strong administrations (with high state capacity) capture, for capital and labour alike, and weak ones lose to flight and evasion. Only the inverse-elasticity burden shift, which acts on *statutory* rates, is specific to the capital account. The effective-rate gradients are therefore predicted under *both* treatments and, as Section 5 shows, found under both, whereas the share gradient is a trade-liberalisation prediction alone.

3 Data, Transformations, and Sample

ETR variables Our panel of effective tax rates is taken from [Bachas et al. \(2023\)](#), who provide the capital effective rate $ETR_K = \tau_K/\theta_K$ and the labour effective rate $ETR_L = \tau_L/\theta_L$ for 156 countries from 1970 to 2018. Here τ_K and τ_L denote capital and labour tax revenue as shares of net domestic product (NDP), and $\theta_K = K\text{-income}/\text{NDP}$ and $\theta_L = L\text{-income}/\text{NDP}$ are the capital and labour income shares of net domestic product, respectively. Tax revenue is constructed from a harmonised concept of direct taxes on capital and labour income, including corporate income tax, withholding taxes, social security contributions on labour, and personal income tax attributed by factor source.

Figure 1 traces the evolution of the two effective rates, with countries split into terciles by their full-period average income per capita and their average state capacity, respectively. It reproduces the central stylised fact of [Bachas et al. \(2023\)](#): since 1990, the capital effective rate has risen in the bottom and middle terciles, the developing world, and fallen in the top, the advanced countries. The labour rate rose across all terciles. These are *unconditional* trends, common to broad groups of countries; whether they reflect the *effect of liberalisation* or a secular force that moves with development is the question the rest of our paper investigates.

But the rise in capital taxation is not confined to the countries that liberalise. Figure 2 plots the same effective rates separately for trade liberalisers, capital-account liberalisers, and the 35 countries that never liberalised on either margin (for definition of liberalisation see below; the full classification of countries by liberalisation status is listed in Appendix A). From the early 1990s onwards, the capital effective rate in panel (a) rises by a similar amount in all three groups, and by slightly more in the control countries; the secular rise is common to

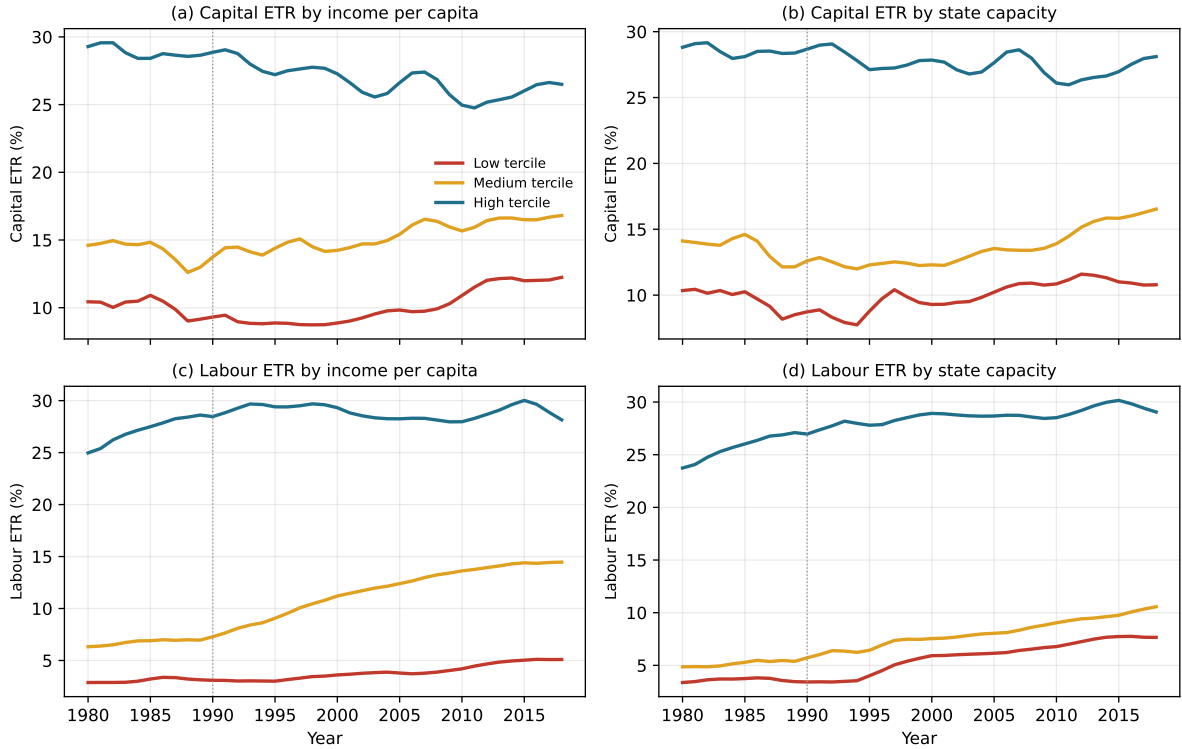


Figure 1: *Stylised facts: effective tax rates by country tercile.* Each panel plots the cross-country mean effective tax rate within terciles of the country's full-period average income per capita (panels a, c) or state capacity (panels b, d), for 1980–2018 (three-year moving average; the dotted vertical line marks 1990).

liberalisers and non-liberalisers alike. This is what a difference-in-differences design exploits: comparing liberalisers with countries never open on the same margin differences away the common trend and isolates the effect of the policy, which we show below is graded by state capacity rather than uniform. Appendix Figure D-4 re-draws these trends with liberalisation-specific controls, re-based to 1990, and shows that control-sample countries commonly grew proportionally *more* than liberalisers.

Factor proxies The two macroeconomic factor proxies are taken from the Penn World Table 11.0 (Feenstra et al. 2015): log real GDP per capita ($\overline{lgdppc}$) and the investment share of GDP ($\overline{inv}$). The third factor proxy is the control-group mean of the outcome itself ($\overline{y^e}$). All these concepts are defined and motivated in Section 4.

Treatment variable Trade openness is measured by the Measure of Aggregate Trade Restrictions (MATR) index of Estefania-Flores et al. (2022), a panel dataset of *trade restrictions* compiled from the IMF's *Annual Report on Exchange Arrangements and Exchange Restrictions*. Higher MATR values indicate more restrictive trade regimes; lower

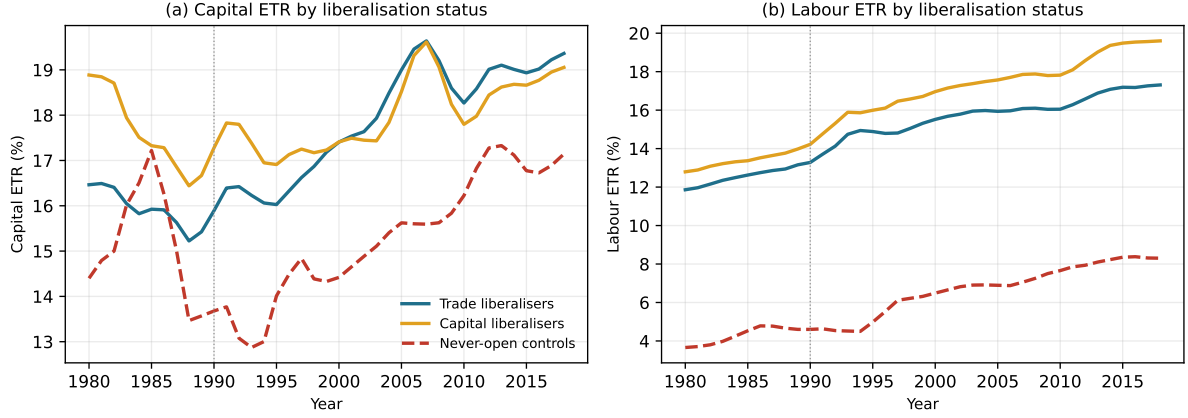


Figure 2: *The rise in effective tax rates is common to liberalisers and non-liberalisers. Cross-country mean effective tax rate for trade liberalisers, capital-account liberalisers, and never-open controls, 1980–2018 (three-year moving average).*

values indicate greater *de jure* openness. Trade-liberalisation is defined as

$$D_{tr,it} = \mathbf{1}[\text{MATR}_{it} < \text{med}(\text{MATR})], \quad (5)$$

where $\text{med}(\text{MATR})$ is the full sample median.

Capital account openness is measured by the Chinn and Ito (2006) index (ka_open), with higher values indicating fewer capital-account restrictions. Capital-account liberalisation is defined as

$$D_{fin,it} = \mathbf{1}[ka_open_{it} > \text{med}(ka_open)]. \quad (6)$$

We use $D_{tr,it}$ and $D_{fin,it}$ as the treatment dummies in the country regressions.

The choice of the pooled sample median as the threshold is conservative: it splits the sample of countries roughly in half at each point in time and ensures that the treatment status is identified from within-country movements rather than from cross-country level differences. This choice is also arbitrary, and in robustness checks (Table 5) we show that our findings are qualitatively unchanged if we use a number of cut-offs between the 40th and 60th percentiles.

State capacity The primary process that moderates the liberalisation-taxation relationship in our analysis is state capacity, taken from O'Reilly and Murphy (2022, O-M). Their index combines Varieties of Democracy (V-Dem) measures of impartial administration, authority over territory, and rule of law. Because this index *excludes* any fiscal or tax indicator, the capacity gradient we estimate cannot be a mechanical reflection of measuring a tax response with a tax-based proxy. The same impartial-administration measure is, across countries, a strong correlate of the tax take (Jensen and Weigel 2025), so the index captures a dimension of the state that independent evidence ties to taxation.

Our baseline results are presented using this state-capacity index, which is constructed from the first principal component of the V-Dem proxies for these measures. A fiscal-based

index from the same source, namely V-Dem’s ‘state fiscal source of revenue’ variable, which captures the administrative-tax margin (Assumption A2) most directly, is used in a robustness check and yields an indistinguishable gradient (Section 5). In both cases, we average the capacity measure over the entire sample period for each country.

State capacity is a slow-moving structural characteristic, but using a time-averaged measure still imposes assumptions about its persistence over time. For one of our main empirical exercises, we split treated countries into terciles, based on their average state capacity index, and in Appendix D.5 we show that almost two-thirds of countries remained in the same capacity tercile over the sample period; only one country moves from the top to the bottom tercile, with another moving in the opposite direction. Studying the annual country rankings from 1980 (ensuring broad country coverage) to 2018, we find that the correlation coefficient with the full-period average we employ is close to 0.9 in every year. In robustness analysis we adopt the capacity average for the 1980s or the index value for the year in which treatment (trade or capital account liberalisation) started, yielding qualitatively identical findings. Because the ranking that defines low- and high-capacity states barely changes, the time-invariant moderator we adopt neither mechanically generates the gradient nor is sensitive to the window over which state capacity is measured.

Sample We define two samples, one for each of the two forms of liberalisation; a country that liberalised on both dimensions appears in both samples. In our baseline result, the *trade sample* comprises countries that have a sustained closed-to-open trade transition (treated) or remain consistently closed throughout the sample (control), excluding countries that are always-open or that switch status more than once. The *capital sample* is constructed analogously for capital-account liberalisation. After applying these criteria and dropping countries with fewer than 25 annual observations, the trade sample contains 85 countries (55 liberalisers and 30 never-open controls, one further control lacking a capacity score) and the capital sample 90 (50 treated and 40 controls); see details in Appendix A. Both samples cover 1970 to 2018. We prune the samples to identify treatment effects in the ‘cleanest’ possible data environment. In robustness checks (Appendix D.7), we reverse or revise the above sample construction rules (we include treatment reverts, set a minimum observation count of 30, 20, or 15) and find qualitatively identical results.

4 Empirical Strategy

Our estimation problem is as follows: each country opens at an individual date, responds to liberalisation in its own way, and is buffeted across 1970 to 2018 by the same global shocks, which affect different economies to different degrees. We wish to quantify each country’s own effect of liberalisation on taxation, clean of those distorting shocks, and we want to know how this effect rises or falls with state capacity. In this section, we build an empirical strategy to address these requirements.

4.1 Econometric model and identification

We take each outcome $y_{f,it} \in \{\text{ETR}_f, \tau_f, \theta_f\}$ to follow a heterogeneous-slope interactive-effects process

$$y_{f,it} = \alpha_{f,i} + \beta_{f,i} D_{it} + \lambda'_{f,i} \mathbf{g}_t + \varepsilon_{f,it}. \quad (7)$$

The term $\mathbf{g}_t$ collects unobserved common factors: forces that move all countries at once, such as global shocks to returns, technology, and tax norms. Each country responds to these forces through its own coefficients $\lambda_{f,i}$, the ‘loadings’, and $\beta_{f,i}$ is the country-specific treatment effect of Section 2. Common factor models are popular in the macro panel literature (e.g. Pesaran 2006, Bai 2009, Chan and Kwok 2022) and have been employed to capture total factor productivity (Eberhardt and Presbitero 2015, Eberhardt and Teal 2020), knowledge spillovers (Eberhardt et al 2013, Ertur and Musolesi 2017), or absorptive capacity (De Visscher et al 2020).

To identify the treatment effect $\beta_{f,i}$, or its average across treated countries, treatment onset must be uncorrelated with the idiosyncratic error once we condition on the factors: $\mathbb{E}[\varepsilon_{f,it} \mid D_{it}, \mathbf{g}_t] = 0$. Our implementation closely resembles Chan and Kwok’s (2022) Principal Component DID (PC-DID) estimator. That estimator first extracts principal components from the residuals of country-specific auxiliary regressions run *in the control sample*. It then adds these components as extra regressors in the treatment regressions of the treated sample, where slopes are free to differ across countries.⁵ Both estimators rest on the same core identification assumption: the factor proxies, whether built from control countries, treated countries, or a mix, must span the space of the unobserved common factors. In plain terms, the proxies have to carry enough information to reconstruct every global shock that matters. This fails if an ‘unspanned’ factor, one the proxies miss, both loads on treated countries and correlates with the timing of liberalisation. The PC-DID tests this with an Alpha test; in Section 5 we introduce the ΔCSD test to do the same in our own implementation. Because we estimate the model one treated country at a time, the usual parallel-trends assumption does not arise here, and by construction we avoid the negative weights (Goodman-Bacon 2021) that can turn a pooled estimate into a misleading average of the individual country effects.

Much of the empirical literature on globalisation and taxation, including Bachas et al. (2023), estimates two-way fixed-effects specifications, $y_{f,it} = \alpha_{f,i} + \beta_f D_{it} + \delta_{f,t} + \varepsilon_{f,it}$. This is a restricted version of (7). It forces a single slope $\beta_{f,i} = \beta_f$ on every country and common loadings $\lambda_{f,i} = \lambda_f$, so that the factor term collapses to simple time effects $\delta_{f,t}$. The pooled $\hat{\beta}_f$ is then read as evidence of how each factor’s fiscal burden evolves with globalisation. Common loadings imply that global shocks (in our sample period, these include the 1970s oil crises, the dot-com bubble, and the Global Financial Crisis) have *identical* impact across countries. Parameter homogeneity is very restrictive, and, indeed, Bachas et al. (2023) adopt a sample selection which assumes limited heterogeneity by focusing on developing countries

⁵Eberhardt (2022) uses the identical setup but replaces the estimated factor proxies in the treatment regressions with cross-section averages à la Pesaran (2006) computed from the control sample only.

only. We relax both restrictions on slopes and loadings: (i) we let $\beta_{f,i}$ vary across countries, and investigate how it varies with state capacity as a mediator (Propositions 1–3), and (ii) we let the loadings be heterogeneous, estimating (7) by the cross-sectional-average augmentation approach (Pesaran 2006) described below. In robustness checks we use income per capita as an alternative mediator, and show that it only imperfectly captures the systematic patterns in the treatment effects that we trace to state capacity. We further estimate a range of alternative implementations (including TWFE and alternative factor models) for our main specification focused on state capacity and find these lead to qualitatively very similar conclusions.

4.2 CCE Estimation

For each liberalising country i in the *trade sample*, we estimate a country-specific treatment regression:

$$y_{it} = \alpha_i + \beta_{tr,i} D_{tr,it} + \lambda'_i \bar{\mathbf{z}}_t + \varepsilon_{it}, \quad (8)$$

where y_{it} is the outcome of interest (e.g. ETR_K , τ_K , θ_K) and $\bar{\mathbf{z}}_t = (\overline{lgdppc}_t, \overline{inv}_t, \bar{y}_t^c)'$ is a vector of cross-sectional averages (CAs). The first two CAs are computed across all 147 countries and proxy global macroeconomic trends in per capita income and investment. The third, $\bar{y}_t^c$, is the cross-sectional average of the outcome (the dependent variable) computed over the never-liberalised control countries only, providing an uncontaminated proxy for the common factor structure that is not driven by treated countries' responses to treatment.⁶ Conceptually, this CA proxy cleanly captures the counterfactual evolution of the outcome as *if the treatment had not happened*, akin to the setup in Chan and Kwok (2022).

We do not include cross-sectional averages of the openness/liberalisation measures *themselves* in the regression (neither of the treatment dummies nor of the underlying openness variables), which would absorb the global liberalisation trend and remove the source of identifying variation. The cross-sectional averages we do include (income, investment, and the control-group outcome) therefore capture the common macroeconomic shocks that would otherwise confound a naïve pooled estimate, while leaving the liberalisation trend itself, our source of identifying variation, in place. In robustness checks we adopt alternative variants of the cross-section averages. Capital-account liberalisation is estimated by the same procedure with $D_{fin,it}$ as the treatment in the *capital sample*.

⁶Estimating each country's effect on its own absorbing closed-to-open transition against never-open controls, and aggregating these effects in the second stage rather than pooling, gives explicit, equal-per-country weights and so avoids the negative-weighting and 'forbidden-comparison' problems that pooled two-way fixed-effects estimators incur under heterogeneous, staggered treatment (Goodman-Bacon 2021, de Chaisemartin and D'Haultfœuille 2020, Callaway and Sant'Anna 2021).

4.3 Continuous and Split Capacity Gradient

To recover the capacity gradient of Propositions 1–3, we regress the country-specific treatment effect $\hat{\beta}_i$ on the country’s average state-capacity $\overline{sc}_i$ (O’Reilly and Murphy 2022):

$$\hat{\beta}_i = \gamma_0 + \gamma_1 \overline{sc}_i + e_i. \quad (9)$$

A positive $\hat{\gamma}_1$ confirms the proposition that the relevant slope is increasing in state capacity. The gradient is estimated over the *treated* countries only, those whose openness status switches during the sample, for which β_i is identified. Control countries, whose treatment dummy never varies, carry no identified treatment effect and are excluded from (9) (they serve only to construct the control-group cross-sectional average $\bar{y}_t^c$).

The second-stage dependent variable $\hat{\beta}_i$ is itself an estimate, and a minority of countries are estimated imprecisely. We deal with this imprecision by estimating (9) with a robust regression M-estimator.⁷ The slope $\hat{\gamma}_1$ captures the marginal change in the treatment effect for a one-unit increase in average state capacity. We report this slope along with the predicted country-specific treatment effect $\hat{\beta}(\overline{sc}) = \hat{\gamma}_0 + \hat{\gamma}_1 \cdot \overline{sc}^*$ evaluated at three points of the cross-country distribution of average state capacity: the mean capacity of the bottom, middle, and top tercile of the pooled treated distribution ($\overline{sc}^* \approx -0.91, 0.44$, and 2.35). These predicted values trace the capacity gradient from low- to high-capacity states and make the model’s predictions (Propositions 1–3) directly testable.

4.4 Cross-sectional Dependence Diagnostics

The relevant identification concern is not whether the per-country residuals exhibit *any* cross-sectional dependence, but whether they exhibit dependence that is *correlated with treatment*. A common factor that loads equally on liberalising and non-liberalising countries, for example, the secular global rise in labour taxation, is benign: it does not bias the estimated treatment effect, because it is differenced away in the comparison of treated and control countries. By contrast, an ‘unspanned’ factor that loads on treated countries and correlates with the timing of liberalisation generates systematic residual structure among treated units that is absent among controls, and biases the estimate.

The Pesaran (2004) CD test does not make this distinction: it rejects whenever residuals exhibit any cross-sectional correlation, benign or not. With $T \approx 45$ it has power to reject mean pairwise correlations below 1%. The same is true of its bias-corrected variant CD* (Pesaran and Xie 2023), designed for residuals from factor-augmented models: it too rejects on any remaining dependence, benign or biasing alike. We therefore use CD* in Section 5 not to

⁷The robust slopes are statistically indistinguishable from those of OLS and of a precision-weighted meta-regression (Lewis and Linzer 2005), because genuine between-country heterogeneity dominates sampling error: the variance of $\hat{\beta}_i$ across countries is roughly eight times its mean sampling variance. A country-cluster two-step bootstrap that resamples countries and re-fits both stages reproduces the analytic standard errors, leaving the high-capacity predicted effect significant at the 5% level; and a leave-one-country-out jackknife confines every gradient to within about 0.2 of its reported value (no single liberaliser drives the result).

test identification but to measure *how much* dependence each estimator leaves in the residual, reported alongside the mean absolute residual correlation. For that purpose the uncorrected CD is unsuitable, since it over-rejects on residuals from which the factor has already been removed.

We use a *differential* diagnostic that targets the identification-relevant alternative directly. Let $\hat{e}_{it}$ be the per-country residual and define the mean absolute pairwise residual correlation within group $g \in \{\text{treated}, \text{control}\}$,

$$\overline{|\rho|}_g = \frac{2}{n_g(n_g - 1)} \sum_{\substack{(i,j) \in g \\ i < j}} |\widehat{\text{corr}}(\hat{e}_{i\cdot}, \hat{e}_{j\cdot})|, \quad (10)$$

where ‘treated’ collects the ever-liberalising countries and ‘control’ the never-liberalising countries in the relevant liberalisation sample. The ΔCSD statistic is

$$\Delta\text{CSD} = \overline{|\rho|}_{\text{treated}} - \overline{|\rho|}_{\text{control}}. \quad (11)$$

Under the null that the cross-sectional averages span the factor space equally for treated and control units, $\mathbb{E}[\Delta\text{CSD}] = 0$; under the alternative that an unspanned factor loads on treated units and correlates with treatment, $\Delta\text{CSD} > 0$. We obtain exact one-sided p -values by permuting the treated/control labels at the country level $B = 999$ times and recomputing ΔCSD , using the $(b + 1)/(B + 1)$ formula of [Phipson and Smyth \(2010\)](#).

4.5 Alternative Empirical Implementations

We demonstrate the robustness of our findings by providing results from a range of alternative estimators, which make different assumptions about the underlying equilibrium relationship. These range from the pooled two-way fixed effects (TWFE) estimator to a naïve Mean Group estimator (MG: [Pesaran and Smith 1995](#)) and two alternative factor-augmented estimators (IFE and PC-DID: [Bai 2009](#), [Chan and Kwok 2022](#)).

For the TWFE we adopt an interaction between the liberalisation dummy D and average state capacity $\overline{sc}$ for each outcome $y_{f,it} \in \{\text{ETR}_f, \tau_f, \theta_f\}$:

$$y_{f,it} = \alpha_{f,i} + \delta_{f,t} + \beta_{0,f}^{\text{TWFE}} D_{it} + \beta_f^{\text{TWFE}} D_{it} \times \overline{sc}_i + \varepsilon_{f,it}, \quad (12)$$

and similarly for the interaction with low, medium, and high state capacity indicators. Here $\beta_{0,f}^{\text{TWFE}}$ is the level effect and β_f^{TWFE} the common capacity slope, so the predicted effect at capacity $\overline{sc}$ is $\beta_{0,f}^{\text{TWFE}} + \beta_f^{\text{TWFE}} \overline{sc}$; standard errors are clustered at the country level.

The Mean Group (MG) and Principal Component Difference-in-Differences (PC-DID)

estimators focus on variants of the implementation for the CCE in equation (8) above:

$$MG: \quad y_{it} = \alpha_i + \beta_{tr,i}^{MG} D_{tr,it} + \eta_i t + \varepsilon_{it}, \quad (13)$$

$$PC-DID: \quad y_{it} = \alpha_i + \beta_{tr,i}^{PC-DID} D_{tr,it} + \lambda_i' \hat{\mathbf{g}}_t + \varepsilon_{it}. \quad (14)$$

The MG crudely models unobserved country-specific heterogeneity with a country-specific linear trend, t . The PC-DID first estimates an auxiliary regression (here, of y on α_i only) in the control sample of countries that never liberalised, then extracts $\hat{r}$ factor proxies $\hat{\mathbf{g}}$ from that regression's residuals by Principal Component Analysis. These factor proxies, with country-specific parameters, then augment the country-specific treatment regressions, instead of the cross-section averages used in the CCE. Aggregation and inference for the MG and PC-DID use the same M-estimation approach as in our CCE implementation; we recover the state-capacity gradient in a separate step from the country estimates $\hat{\beta}_i$ and $\overline{sc}_i$ (or the three values for $\overline{sc}^*$).

Finally, Bai's (2009) interactive fixed-effects (IFE) estimator works much like the PC-DID, but it extracts the principal components from the pooled regression of treated countries (following Gobillon and Magnac 2016) and specifies an interaction between liberalisation D and state capacity $\overline{sc}_i$:

$$y_{f,it} = \alpha_{f,i} + \delta_{f,t} + \lambda_{f,i}' \mathbf{f}_t + \beta_{0,f}^{IFE} D_{it} + \beta_f^{IFE} D_{it} \times \overline{sc}_i + \varepsilon_{f,it}. \quad (15)$$

The interactive fixed-effects term $\lambda_{f,i}' \mathbf{f}_t$ collects $\hat{r}$ common factors estimated by principal components (Bai 2009), augmenting the additive country and time effects. Although the IFE estimates a common treatment effect, it is robust to treatment-effect heterogeneity. The number of relevant factors $\hat{r}$ in the PC-DID and IFE implementations is established using Ahn and Horenstein's (2013) eigenvalue-ratio. Together, these five estimators allow us explore whether the capacity gradient is a feature of the data or just an artefact of the assumptions of one single estimator.

5 Results

5.1 Pooled Correlations with *de facto* Openness

We begin by establishing that our data reproduce the core correlations in Bachas et al. (2023), implying that capital taxation *rose* with globalisation in the developing world, using *de facto* measures for globalisation. We regress the effective tax rate for capital or labour on openness using standard two-way fixed effects estimators on the full sample of 147 countries, reporting the pooled coefficient and its split (determined by the cross-country median of income per capita) between developed and developing countries; see Table 1. For trade openness, and for capital openness (whether adopting total external assets and liabilities or FDI alone), the

capital effective rate rises with openness and the rise sits in the developing country group (+2.7, +2.1 and +1.6 points per standard deviation, all statistically significant), while the developed-country effect is small and insignificant. These developing-country effects survive country-specific linear trends for trade openness as well as capital openness measured via FDI in column (2). The positive trade liberalisation effect is also present in the ETR for labour, again robust to the inclusion of country trends, columns (3) and (4). Having thus established the benchmark results from [Bachas et al. \(2023, Table 1, column \(1\)\)](#), we now turn to our own research design, based on liberalisation *episodes* rather than *de facto* volumes.

Table 1: Correlations of *de facto* openness with the effective tax rate.

	ETR _K		ETR _L	
	TWFE	+trend	TWFE	+trend
Trade openness, log	+1.98** (0.78)	+1.05 (0.65)	+0.95** (0.41)	+0.68*** (0.25)
developed	+0.37 (1.27)	−1.04 (1.77)	+1.50* (0.86)	−0.08 (0.53)
developing	+2.71*** (0.88)	+1.58** (0.64)	+0.70** (0.34)	+0.87*** (0.25)
Capital openness, ln(IFI/GDP)	+1.44** (0.68)	+0.39 (0.68)	+1.12** (0.44)	−0.57** (0.26)
developed	+1.05 (0.88)	−0.56 (1.29)	+1.98*** (0.55)	−0.44 (0.75)
developing	+2.13*** (0.77)	+0.77 (0.75)	−0.41 (0.34)	−0.62** (0.27)
Capital openness, ln(FDI/GDP)	+1.18 (0.76)	−0.22 (0.88)	+0.71* (0.37)	+0.61* (0.32)
developed	+0.55 (1.04)	−3.02* (1.80)	+1.68*** (0.54)	+0.15 (0.67)
developing	+1.57** (0.76)	+1.08* (0.58)	+0.11 (0.32)	+0.82*** (0.26)

Notes: We report estimates from standard two-way fixed-effects regressions of the effective tax rate on the indicated openness measure, full sample of 147 countries, adopting country-clustered standard errors (reported in parentheses). Coefficients indicate the effect of a standard deviation increase in the measure. For each measure we report a full sample estimate and estimates for developed and developing countries, respectively, split at the cross-country median of income per capita (Penn World Table real GDP per capita); the “+trend” column adds country-specific linear trends. Trade openness is $(X+M)/GDP$ from the Penn World Table, in logarithms; the two capital-openness measures are from the External Wealth of Nations database ([Lane and Milesi-Ferretti 2018](#)): *total* international financial integration (IFI), all foreign assets plus liabilities over GDP, and *FDI openness*, inward plus outward direct-investment stocks over GDP; both in logarithms. Statistical significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5.2 Fiscal gradients

Table 2 reports, for all decomposition outcomes (ETR, tax revenue τ , and factor share θ for capital and labour, respectively) and both forms of liberalisation, (i) the robust mean of the treatment effect (ATET) $\hat{\beta}$ ignoring state capacity, (ii) the robust-regression gradient

slope $\hat{\gamma}_1$ of the per-country effect on average state capacity, (iii) the predicted treatment effect $\hat{\beta} = \hat{\gamma}_0 + \hat{\gamma}_1 \overline{sc}^*$ at the bottom, middle, and top tercile of the pooled treated capacity distribution, and (iv) the Δ CSD identification diagnostic. These are the results based on the CCE estimator in equation (8). The predicted-effect format (low/median/high capacity) is shared across Tables 2–5. Figure 3 traces the state capacity gradient nonparametrically.

Diagnostics Before interpreting the estimates we probe the identifying assumption from several directions. The Δ CSD permutation test passes for every result in Table 2: the permutation p -values range from 0.61 to 1.00, and the Δ CSD point estimates are negative and near-zero throughout, so treated residuals carry no more cross-sectional dependence than control residuals. This test is a necessary check rather than a sufficient one. As Appendix E shows by simulation, it has power against an unspanned factor that the estimator leaves in the residual, or whose timing is misaligned with the onsets, but is weak against a factor that moves *with* the onset itself, because the treatment dummy then absorbs that factor out of the treated residual, contaminating $\hat{\beta}_{f,i}$ while leaving the residual looking clean. Because treatment onsets are staggered, this gap is narrow: a common factor cannot move with every country’s onset, so only a confounder timed to *each* country’s individual opening up escapes the net. We therefore do not rest identification on our CCE estimator alone. The capacity gradient reproduces across five estimators with different factor treatments (Table 3), survives a random-capacity placebo (below), and, under a date-randomisation placebo, concentrates at the true liberalisation dates rather than a pre-existing capacity-graded trend (Appendix F). The leading concrete candidate for a coincident, capacity-correlated confounder is a value-added-tax reform bundled with liberalisation; Appendix C.3 shows that the capacity-graded movement in consumption taxes such a reform would require is absent from the data.

In levels, the CCE residuals carry about half the cross-sectional dependence that a two-way fixed-effects model leaves on the same sample (a mean absolute pairwise residual correlation near 0.24 against 0.43), and are equally dependent across treated and control units, the signature of common factors that have been spanned rather than left in the error.⁸

Section 5.3 reports the residual-dependence for all five estimators we use for robustness (Table 4). This tests precisely the conditional-exogeneity of Section 4: the cross-sectional

⁸The residual level itself is benign, on two counts. First, the figure is a mean *absolute* pairwise correlation, which stays well above zero even under cross-sectional independence: with $T \approx 45$ annual observations, the sampling distribution of the correlation between two independent series has standard deviation $\approx 1/\sqrt{T}$, so its expected absolute value is about 0.12. Permuting each control country’s residuals in time independently, which destroys any cross-country co-movement while preserving each series’ length and gaps, reproduces this floor at 0.14. The CCE residuals sit at 0.24, about 0.10 above the independence floor, and their mean *signed* correlation is ≈ 0.02 : what remains is sign-incoherent weak dependence, not a coherent common factor, which would pull the signed mean up towards the absolute one. Second, that residue does not distort inference. A block bootstrap robust to cross-sectional dependence leaves the gradient standard errors essentially unchanged. It resamples four-year blocks of time while holding the full country cross-section fixed, so any residual co-movement passes into the sampling distribution of the gradient. The bootstrap-to-analytic ratio is 0.93 and 0.98 on the two capital effective-rate estimates and at most 1.24 on the labour one, and every capacity gradient stays statistically significant. Residual cross-sectional dependence at this level therefore threatens neither the identification of the per-country effects nor the precision of the gradient.

averages span the common factor structure equally for treated and control units, so the per-country effects are not contaminated by an unspanned, treatment-correlated factor.

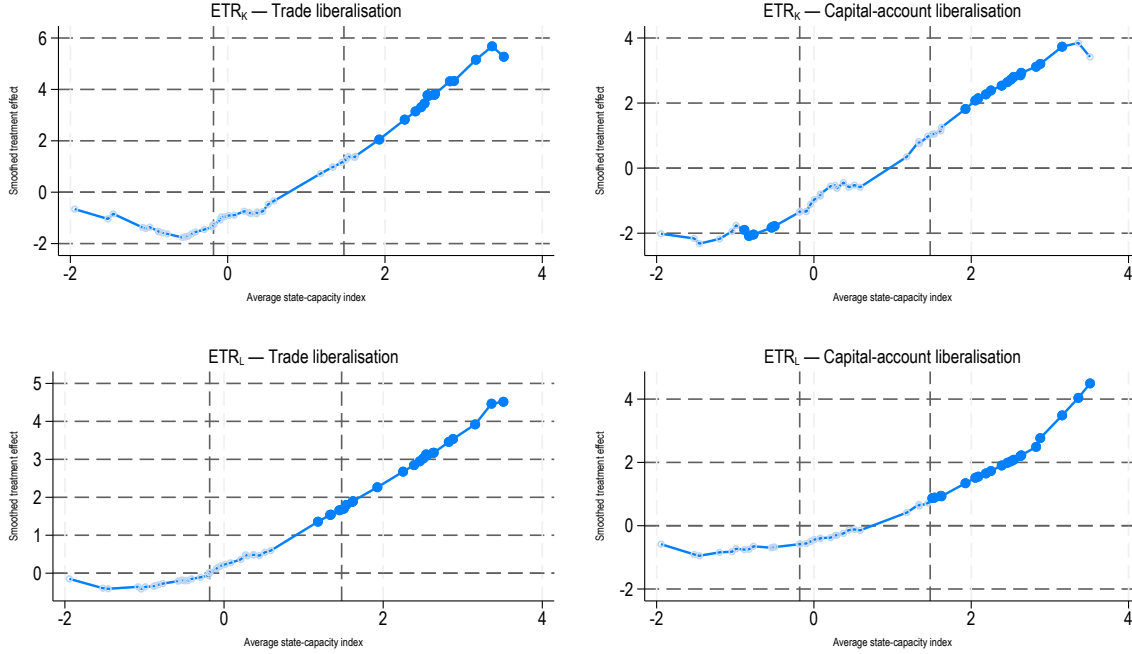


Figure 3: The state-capacity gradient of globalisation's effect on factor taxation. Each panel plots a running-line smoother of the country-specific CCE treatment effect $\hat{\beta}_i$ for the indicated effective tax rate against the country's average state-capacity index $\overline{sc}_i$ (O'Reilly and Murphy 2022; treated countries only); the two dashed vertical lines mark the cut-offs separating the bottom, middle, and top terciles of state capacity. Solid (hollow) markers on the running line are (in)significantly different from zero at the 10% level; country markers are minimally perturbed for ease of presentation.

Average Treatment Effects ignoring state capacity Before turning to our findings relating liberalisation and state capacity, we briefly demonstrate the uninformative nature of the average treatment effect (ATET) of liberalisation on taxation measures, based on the robust means of the $\hat{\beta}_i$ of equation (8), reported in the top row of each panel of Table 2. For the capital effective rate the average effect is small and insignificant (+0.1 under trade, +0.3 under capital-account liberalisation), even though (looking ahead) the capacity gradient beneath will turn out to be strong: the average nets the negative low-capacity effect against the positive high-capacity one and so conveys neither. The labour rate will be shown to rise more uniformly across capacity, so its trade average is positive.

Main Findings (ETR , τ) The pattern is the same across all tax outcomes and is summarised by the predicted effect at low, median, and high state capacity (Table 2). Consider *trade liberalisation* first. The capacity gradients of capital and labour tax revenue

Table 2: Average treatment effects and the state-capacity (SC) gradient.

	Capital			Labour		
	(1) ETR _K	(2) τ_K	(3) θ_K	(4) ETR _L	(5) τ_L	(6) θ_L
Trade liberalisation (D_{tr})						
ATET $\hat{\beta}$ (ignoring SC)	+0.09 (0.74)	+0.25 (0.22)	+0.87* (0.51)	+0.93*** (0.35)	+0.42** (0.20)	-0.87* (0.51)
SC Gradient $\hat{\gamma}_1$	+1.42*** (0.54)	+0.47*** (0.13)	-0.35 (0.45)	+0.94*** (0.23)	+0.41*** (0.16)	+0.35 (0.45)
$\hat{\beta}$ (low SC)	-1.97** (0.82)	-0.49* (0.26)	+1.35** (0.67)	-0.34 (0.41)	-0.17 (0.24)	-1.35** (0.67)
$\hat{\beta}$ (medium SC)	-0.06 (0.69)	+0.15 (0.20)	+0.88* (0.51)	+0.93*** (0.34)	+0.39* (0.21)	-0.88* (0.51)
$\hat{\beta}$ (high SC)	+2.66* (1.42)	+1.04*** (0.32)	+0.21 (1.12)	+2.71*** (0.61)	+1.18*** (0.43)	-0.21 (1.12)
Δ CSD	-0.034	-0.005	-0.066	-0.006	-0.016	-0.066
p -value	0.985	0.612	0.997	0.636	0.829	0.999
N_{tr}	55	55	55	55	55	55
N_c	31	31	31	31	31	31
Capital-account liberalisation (D_{fin})						
ATET $\hat{\beta}$ (ignoring SC)	+0.28 (0.79)	+0.30 (0.26)	+1.01* (0.55)	+0.51 (0.32)	+0.15 (0.20)	-1.01* (0.55)
SC Gradient $\hat{\gamma}_1$	+1.34*** (0.47)	+0.56*** (0.14)	+0.37 (0.38)	+0.70*** (0.24)	+0.33* (0.20)	-0.37 (0.38)
$\hat{\beta}$ (low SC)	-2.05** (1.03)	-0.67** (0.33)	+0.28 (1.06)	-0.59 (0.46)	-0.36 (0.28)	-0.28 (1.06)
$\hat{\beta}$ (medium SC)	-0.24 (0.76)	+0.09 (0.23)	+0.79 (0.67)	+0.35 (0.31)	+0.08 (0.21)	-0.79 (0.67)
$\hat{\beta}$ (high SC)	+2.32** (1.13)	+1.16*** (0.33)	+1.50** (0.64)	+1.68*** (0.55)	+0.71 (0.48)	-1.50** (0.64)
Δ CSD	-0.041	-0.023	-0.044	-0.013	-0.018	-0.044
p -value	0.992	0.903	0.977	0.773	0.852	0.977
N_{tr}	50	50	50	50	50	50
N_c	40	40	40	40	40	40

Notes: Standard errors in parentheses. We report results based on country-specific regressions $y_{it} = \alpha_i + \beta_i D_{it} + \lambda_i' \bar{z}_t + \varepsilon_{it}$, with $\bar{z}_t$ as defined in Section 4, estimated over treated countries N_{tr} only. Control sample size N_c is reported for reference. Treatment refers to trade D_{tr} or capital account liberalisation D_{fin} , as defined in Section 3. Outcomes y are ETR, the effective tax rate; τ , tax revenue; and θ , the factor income share (all for K capital and L labour, respectively). For the ATET the $\hat{\beta}_i$ are regressed on a constant term by robust (Huber M-)regression. Mean Group estimates reported use the $\hat{\beta}_i$ regressed on the country's average state-capacity (SC) index (O'Reilly and Murphy 2022) by robust (Huber M-)regression. $\hat{\gamma}_1$ is the gradient slope; $\hat{\beta}$ is the predicted effect at the *mean capacity of each tercile* of the pooled treated distribution (the average bottom/middle/top-third liberaliser): low ≈ -0.9 , medium ≈ 0.4 , high ≈ 2.4 . Δ CSD is the treated-minus-control difference in mean absolute pairwise residual correlation [equation (11)], with p -values constructed using permutation. Statistical significance on $\hat{\gamma}_1$ and $\hat{\beta}$: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

(τ_K and τ_L) are both positive and highly significant. The predicted effect on τ_K runs from a significant contraction of about 0.5 percentage points of NDP in low-capacity states to a significant expansion of about 1.0 in high-capacity ones; the effect on τ_L follows the same path, from an insignificant -0.2 at low capacity to a significant $+1.2$ at high. The two effective rates show the same gradient: the predicted effect on ETR_K moves from a significant -2.0 percentage points in low-capacity states to $+2.7$ in high-capacity ones, while on ETR_L it runs from -0.3 (insignificant at the representative low-capacity country) to a significant $+2.7$. Across liberalisers, a country's capital and labour responses are positively correlated (correlation $+0.5$ under trade liberalisation): in most, opening moves the two in the same direction, raising both where capacity is high, and eroding the capital base where it is low. *Capital-account liberalisation* delivers the same picture: the predicted ETR_K and τ_K effects run from a significant negative value in low-capacity states to a positive value in high-capacity ones, crossing zero near the median. The equivalent effects for labour (ETR_L and τ_L) are less precisely estimated but follow identical patterns. Relative to an income-based gradient (see Table 6), the state-capacity gradient is uniformly larger and statistically significant at the 1% level throughout (with the exception of τ_L for capital-account liberalisation: 10% level).⁹ Figure 3 traces the same sign change nonparametrically and leaves little doubt that only countries in the top capacity tercile had positive significant treatment effects.

These results support Propositions 2 and 3. Trade liberalisation or capital-account opening raises the captured capital tax base where administrative capacity is high and erodes it where capacity is low (Proposition 2), so the effect on τ_K and ETR_K is increasing in state capacity. The effective taxation of labour also rises with openness, and more so in higher-capacity economies: the same administrative capacity that captures capital income captures labour income too (Proposition 3), under both forms of liberalisation. The gradient on labour tax revenue τ_L is positive but smaller than that on the labour ETR, because the higher effective rate is partly offset by the falling labour income share (A1). Our discussion in Section 5.4 will uncover that foreign direct investment (FDI) is the driving force behind these results.

Factor reallocation (θ) Proposition 1 predicts that trade liberalisation shifts income towards capital, more so with state capacity. However, our estimates validate at most half of it: the average effect on the capital share is positive but only marginally significant ($+0.87$, at 10%), and the state-capacity gradient is negative, though too imprecise to sign. We locate our robust evidence in the tax outcomes of the previous subsection. The contrast is itself informative: the development-graded effect of globalisation operates through the statutory-rate and enforcement components (κ_f, ϕ_f , whose product is the effective rate $ETR_f = \kappa_f \phi_f$ of equation (1)), which we estimate precisely, rather than through factor reallocation (θ_f), which we do not.

⁹The gradient is robust to minimum-observations and permanence rules, to measuring capacity in the 1980s or at the onset rather than averaged over the full period, to the choice of cross-sectional averages, and to replacing the capacity index with the [Hanson and Sigman \(2021\)](#) alternative (Appendix D). We consider alternative estimators and liberalisation cutoffs in Section 5.3.

5.3 Robustness

Alternative implementations The conditional, state capacity-graded nature of the effect, and its robustness across alternative estimators, is displayed in Table 3, which reports the *predicted treatment effect* on the two effective rates at the bottom, middle, and top tercile of state capacity, from five capacity-varying estimators, all on the same samples of trade or capital-account liberalisers and never-open controls: a two-way fixed-effects model with a capacity interaction, a mean-group estimator (MG, *without* cross-sectional averages), our CCE estimator, and two further estimators that strip out the common factors a different way, the Bai (2009) interactive-fixed-effects (IFE) estimator,¹⁰ and the principal-components difference-in-differences (PC-DID) estimator of Chan and Kwok (2022).¹¹

The estimators tell the same story for capital, under both trade and capital-account liberalisation: the effect is negative or insignificant in low-capacity states and rises to a positive, significant effect in high-capacity states (CCE results are those reported in columns (1) and (4) of Table 2). The positive, significant high-capacity capital effect from our CCE benchmark is reproduced by three of the four alternative estimators, for both forms of liberalisation; under IFE the trade effect is positive but insignificant (+1.22). The labour effective rate shows the same upward gradient, with a high-capacity labour effect that is positive and significant under every estimator for each form of liberalisation. The fact that this pattern prevails, whether one removes common factors, models heterogeneity, or pools, confirms it is a feature of the data, not of the estimator. It also makes plain why a homogeneous, pooled coefficient ignoring state capacity is uninformative about the mechanism (the ATET row of Table 2).

Diagnostics for robustness specifications Table 4 reports the residual cross-sectional dependence for all five estimators. Panel A gives the level of dependence among the never-open controls, a measure of how much of the common factor each estimator leaves in the error. The two estimators that do not model the factors leave a mean absolute pairwise residual correlation between 0.34 and 0.57: the mean-group estimator, which ignores the factor structure, and two-way fixed effects, which removes only additive time effects. The three factor estimators almost halve it, to between 0.23 and 0.32, and they do so regardless of how they handle the factors: through cross-sectional averages (CCE), principal components inside

¹⁰The number of common factors is selected by the eigenvalue-ratio test of Ahn and Horenstein (2013) (two for the capital effective rate, one for the labour rate); we follow Gobillon and Magnac (2016) in applying the estimator to the difference-in-differences design.

¹¹As a reminder: TWFE $\times \overline{sc}_i$ and IFE pool one state capacity interaction *across all countries*, whereas the MG, CCE and PC-DID estimators fit a separate effect for each country and average it, allowing for the heterogeneity across treated countries. They also handle the common factors differently: the MG estimator does not model them (a no-control benchmark); TWFE remove only additive time effects, valid only if all countries share the same factor loadings, which their higher residual dependence rejects (Table 4); CCE proxies the factors with cross-sectional averages (Pesaran 2006); IFE estimates them by principal components within the regression; and PC-DID enters principal-component proxies as per-country control functions. For CCE and PC-DID the common component is identified from the never-open controls, not from the treated countries' own response.

Table 3: Treatment effect by state-capacity level, various estimators.

	(1) TWFE	(2) MG	(3) CCE	(4) IFE	(5) PCDID
Trade liberalisation (D_{tr}): ETR_K					
SC Gradient $\hat{\gamma}_1$	+1.67*** (0.54)	+1.52** (0.64)	+1.42*** (0.54)	+1.37*** (0.46)	+1.84*** (0.61)
$\hat{\beta}$ (low SC)	-1.14 (1.65)	-0.03 (0.97)	-1.97** (0.82)	-3.25*** (0.99)	+0.02 (0.84)
$\hat{\beta}$ (medium SC)	+1.11 (1.47)	+2.02** (0.90)	-0.06 (0.69)	-1.40 (0.87)	+2.50*** (0.70)
$\hat{\beta}$ (high SC)	+4.30** (1.78)	+4.93*** (1.78)	+2.66* (1.42)	+1.22 (1.32)	+6.01*** (1.58)
Trade liberalisation (D_{tr}): ETR_L					
SC Gradient $\hat{\gamma}_1$	+1.43*** (0.40)	+1.52*** (0.33)	+0.94*** (0.23)	+0.79*** (0.27)	+1.56*** (0.30)
$\hat{\beta}$ (low SC)	-1.29* (0.74)	+1.58*** (0.44)	-0.34 (0.41)	-0.86** (0.40)	-0.10 (0.44)
$\hat{\beta}$ (medium SC)	+0.64 (0.67)	+3.63*** (0.50)	+0.93*** (0.34)	+0.20 (0.33)	+1.99*** (0.42)
$\hat{\beta}$ (high SC)	+3.37*** (1.14)	+6.53*** (1.01)	+2.71*** (0.61)	+1.70** (0.71)	+4.97*** (0.84)
Countries	85	55	55	85	55
Capital-account liberalisation (D_{fin}): ETR_K					
SC Gradient $\hat{\gamma}_1$	+1.21** (0.60)	+0.95 (0.66)	+1.34*** (0.47)	+1.37*** (0.43)	+1.88*** (0.69)
$\hat{\beta}$ (low SC)	-0.75 (1.66)	+0.28 (1.06)	-2.05** (1.03)	-1.70* (0.92)	-0.51 (1.03)
$\hat{\beta}$ (medium SC)	+0.88 (1.40)	+1.56 (1.05)	-0.24 (0.76)	+0.15 (0.76)	+2.02** (0.91)
$\hat{\beta}$ (high SC)	+3.20* (1.75)	+3.38* (1.95)	+2.32** (1.13)	+2.77** (1.15)	+5.61*** (1.86)
Capital-account liberalisation (D_{fin}): ETR_L					
SC Gradient $\hat{\gamma}_1$	+1.05*** (0.36)	+1.07*** (0.40)	+0.70*** (0.24)	+0.85** (0.37)	+0.94*** (0.32)
$\hat{\beta}$ (low SC)	-0.34 (0.71)	+2.01*** (0.46)	-0.59 (0.46)	-0.76 (0.62)	-0.20 (0.45)
$\hat{\beta}$ (medium SC)	+1.07* (0.63)	+3.45*** (0.56)	+0.35 (0.31)	+0.38 (0.44)	+1.07*** (0.38)
$\hat{\beta}$ (high SC)	+3.07*** (1.02)	+5.49*** (1.21)	+1.68*** (0.55)	+2.01** (0.90)	+2.87*** (0.85)
Countries	90	50	50	90	50

Notes: Standard errors in parentheses. ETR results using alternative implementations. TWFE – two-way FE model with liberalisation dummy interacted with average state capacity, country-clustered SE. MG – mean-group estimator. CCE – MG augmented with CAs. IFE – Bai (2009) interactive fixed effects, treatment interacted with capacity, at the eigenvalue-ratio factor count (two factors for ETR_K , one for ETR_L). PC-DID – Chan and Kwok (2022) principal-components DID estimator. MG, CCE, and PC-DID aggregated using the same robust M-estimator as the baseline and predicted at the same tercile centroids; TWFE and IFE estimate the capacity interaction directly. Statistical significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. See Table 2 for all other details.

Table 4: Residual cross-sectional dependence by estimator.

Estimator	Trade lib.		Capital-acc. lib.	
	(1) ETR _K	(2) ETR _L	(3) ETR _K	(4) ETR _L
<i>Panel A. Residual CSD, control group (level)</i>				
TWFE $\times$ cap.	0.34	0.57	0.34	0.48
MG, no CA	0.35	0.51	0.35	0.50
CCE	0.23	0.24	0.24	0.24
IFE	0.26	0.30	0.24	0.32
PC-DID	0.24	0.24	0.23	0.26
<i>Panel B. ΔCSD permutation p-value</i>				
TWFE $\times$ cap.	0.12	0.99	0.10	0.66
MG, no CA	0.99	1.00	1.00	1.00
CCE	0.99	0.63	1.00	0.78
IFE	0.71	0.20	0.47	0.48
PC-DID	0.58	0.47	0.85	0.65
<i>Panel C. CD* statistic (bias-corrected, full panel)</i>				
TWFE $\times$ cap.	-3.40***	+7.57***	-2.92***	+0.61
MG, no CA	+30.52***	+2.93***	+3.42***	+14.34***
CCE	+2.14**	+6.24***	+3.15***	+6.65***
IFE	+0.62	+5.83***	-2.09**	+2.94***
PC-DID	+15.34***	+3.56***	+9.43***	+4.56***

Notes: Cross-sectional dependence of the estimated residuals (the object of Section 4.4), by estimator and cell, on the same clean sample of liberalisers and never-open controls as Table 3. *Panel A* reports the level, the mean absolute pairwise residual correlation among the never-open control countries; a lower value means less of the common factor is left in the error. *Panel B* reports the permutation p -value of the Δ CSD = $|\bar{\rho}|_{\text{treated}} - |\bar{\rho}|_{\text{control}}$ test [equation (11)]: holding the residuals fixed, the treated and control labels are permuted at the country level 999 times, and the one-sided p -value is the share of permutations whose Δ CSD is at least the observed value. A small p indicates a statistically significant treatment-correlated component of the residual dependence. *Panel C* reports the bias-corrected CD* statistic of Pesaran and Xie (2023), computed over all countries in the cell; it is standard normal under the null of cross-sectional independence, and a significant value indicates that residual dependence remains, of either sign. TWFE and MG do not model the common factor; CCE, IFE, and PC-DID do. Residuals for IFE are taken from the interactive-fixed-effects fits underlying Table 3; the other estimators are computed directly. Statistical significance in Panel C: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (two-sided).

the regression (IFE), or via control functions (PC-DID). The residual cleaning is therefore a property of factor modelling itself, not of our particular implementation.

Panel B turns to the significance of the error dependence: it reports the permutation p -value of the Δ CSD test in equation (11), the one-sided probability that treated residuals are no more cross-sectionally dependent than control residuals, so a small value signals a statistically significant treatment-correlated component. CCE, PC-DID and IFE comfortably pass this test in every specification, with no p -value below 0.17: the cross-sectional averages and the estimated factors leave no detectable treatment-correlated dependence. Two-way fixed effects is the only estimator the test flags, and it does so in the two capital effective rate specifications ($p = 0.08$ under capital-account liberalisation, 0.11 under trade), the outcome carrying our central result that the capital ETR rises with state capacity; a pooled coefficient would inherit the treatment-correlated dependence the test detects. The diagnostics for the MG estimator

call for a caveat: its high p -value does not indicate clean residuals. The treatment dummy absorbs the common treatment-timing movement in the treated units, so their residuals look less dependent even though, as Panel A shows, the factor is still in the error.

Panel C reports the bias-corrected CD^* statistic of [Pesaran and Xie \(2023\)](#), which tests the stricter null hypothesis that the residuals are merely cross-sectionally weakly dependent in levels, rather than the treated-minus-control comparison of Panel B. It rejects for almost every estimator and specification, the three factor estimators included. This is the feature already noted in Section 4.4: with $N \approx 86$ countries and $T \approx 45$ years the test has power against mean correlations below 1%, so it rejects even the weak, sign-incoherent residue the factor estimators leave (the CCE residuals' *signed* mean pairwise correlation is only ≈ 0.00 , as noted above). A significant CD^* therefore says only that the residual dependence is not exactly zero, not that it is large or that it biases the gradient. Whether it biases the gradient is the question Panel B answers: ΔCSD isolates the component of the dependence that is *correlated with treatment*, the only component that does. A common factor loading equally on liberalisers and controls inflates CD^* while leaving ΔCSD at zero, exactly the configuration the factor estimators display. Using simulation, Appendix E confirms, at this panel's dimensions, that ΔCSD holds its size under such a benign factor and gains power against a treatment-correlated one, while the raw CD over-rejects on defactored residuals, the reason Panel C reports the bias-corrected CD^* .

Alternative definitions of liberalisation The conditional pattern is also robust to the cutoff used to define opening up. Our dummies set the open/closed threshold at the pooled median of the respective openness index; Table 5 re-estimates the predicted impact at the bottom, middle, and top tercile of state capacity as the cutoff varies from the 40th to the 60th percentile, rebuilding the dummies, the treated/control split, and the control cross-sectional average at each cutoff. The low-capacity impact is negative and the high-capacity impact positive at every cutoff; the median impact is near zero for the capital rate and modestly positive for labour. The significantly negative low-capacity effect on ETR_K , the most pronounced feature of the capacity gradient, survives almost every cutoff for both liberalisations (p45 for trade is the exception).

Alternative definitions of state capacity State capacity might be a consequence of opening up rather than a cause of the gradient. Trade liberalisation can itself *build* it, as states replace lost tariff revenue by developing domestic tax administration ([Baunsgaard and Keen 2010](#)). The full-period capacity average would then partly post-date treatment. We address this by re-estimating the gradient on each country's *pre-onset* capacity, measured entirely before its liberalisation, and using the fiscal sub-index, the component most exposed to this concern since liberalisation most plausibly builds tax administration. The gradient attenuates modestly but remains positive and significant for every outcome: under trade liberalisation $\hat{\gamma}_1 = 1.09$ ($t = 2.6$) for ETR_K and 0.81 ($t = 3.8$) for ETR_L ; under capital-

Table 5: Predicted impact at low/medium/high state capacity, by liberalisation cutoff.

	$p40$	$p45$	$p50$	$p55$	$p60$
Trade liberalisation (D_{tr})					
$ETR_K, \hat{\beta}$ low SC	-1.50* (0.89)	-1.40 (0.91)	-1.97** (0.82)	-1.62* (0.98)	-1.63* (0.97)
$\hat{\beta}$ medium SC	-0.34 (0.66)	-0.12 (0.74)	-0.06 (0.69)	-0.14 (0.71)	-0.07 (0.71)
$\hat{\beta}$ high SC	+1.31 (1.34)	+1.69 (1.44)	+2.66* (1.42)	+1.96* (1.19)	+2.15* (1.22)
$\tau_K, \hat{\beta}$ low SC	-0.59* (0.31)	-0.46 (0.31)	-0.49* (0.26)	-0.31 (0.26)	-0.33 (0.27)
$\hat{\beta}$ medium SC	+0.08 (0.21)	+0.15 (0.22)	+0.15 (0.20)	+0.02 (0.20)	+0.03 (0.21)
$\hat{\beta}$ high SC	+1.02*** (0.25)	+1.02*** (0.29)	+1.04*** (0.32)	+0.49 (0.32)	+0.54 (0.33)
$ETR_L, \hat{\beta}$ low SC	-0.39 (0.53)	-0.10 (0.41)	-0.34 (0.41)	-0.28 (0.32)	-0.26 (0.31)
$\hat{\beta}$ medium SC	+0.82** (0.38)	+1.29*** (0.36)	+0.93*** (0.34)	+0.91*** (0.32)	+0.89*** (0.33)
$\hat{\beta}$ high SC	+2.52*** (0.73)	+3.27*** (0.78)	+2.71*** (0.61)	+2.59*** (0.63)	+2.51*** (0.75)
$\tau_L, \hat{\beta}$ low SC	-0.22 (0.28)	-0.04 (0.23)	-0.17 (0.24)	-0.19 (0.21)	-0.19 (0.21)
$\hat{\beta}$ medium SC	+0.44* (0.25)	+0.70*** (0.21)	+0.39* (0.21)	+0.61** (0.24)	+0.61*** (0.23)
$\hat{\beta}$ high SC	+1.38** (0.58)	+1.75*** (0.45)	+1.18*** (0.43)	+1.73*** (0.53)	+1.75*** (0.47)
Capital-account liberalisation (D_{fin})					
$ETR_K, \hat{\beta}$ low SC	-1.86* (1.00)	-1.93* (1.03)	-2.05** (1.03)	-1.73* (0.89)	-2.00** (0.80)
$\hat{\beta}$ medium SC	-0.34 (0.76)	-0.27 (0.76)	-0.24 (0.76)	-0.21 (0.70)	-0.05 (0.68)
$\hat{\beta}$ high SC	+1.80* (1.08)	+2.07* (1.09)	+2.32** (1.13)	+1.95 (1.26)	+2.71** (1.29)
$\tau_K, \hat{\beta}$ low SC	-0.57* (0.33)	-0.68** (0.33)	-0.67** (0.33)	-0.46* (0.26)	-0.51** (0.26)
$\hat{\beta}$ medium SC	+0.06 (0.23)	+0.09 (0.24)	+0.09 (0.23)	-0.04 (0.22)	-0.01 (0.22)
$\hat{\beta}$ high SC	+0.94*** (0.26)	+1.18*** (0.31)	+1.16*** (0.33)	+0.55 (0.35)	+0.70* (0.37)
$ETR_L, \hat{\beta}$ low SC	-0.37 (0.44)	-0.62 (0.44)	-0.59 (0.46)	-0.34 (0.31)	-0.29 (0.28)
$\hat{\beta}$ medium SC	+0.71** (0.32)	+0.34 (0.30)	+0.35 (0.31)	+0.94** (0.38)	+0.81** (0.40)
$\hat{\beta}$ high SC	+2.22*** (0.58)	+1.71*** (0.53)	+1.68*** (0.55)	+2.75*** (0.95)	+2.36** (0.99)
$\tau_L, \hat{\beta}$ low SC	-0.30 (0.28)	-0.36 (0.26)	-0.36 (0.28)	-0.21 (0.21)	-0.18 (0.18)
$\hat{\beta}$ medium SC	+0.38 (0.23)	+0.11 (0.19)	+0.08 (0.21)	+0.60** (0.30)	+0.56** (0.25)
$\hat{\beta}$ high SC	+1.34** (0.55)	+0.76* (0.44)	+0.71 (0.48)	+1.76** (0.80)	+1.61** (0.65)

Notes: Standard errors in parentheses. We present CCE-MG based predictions for ETR and τ at the low, medium, and high state capacity (SC) terciles as in Table 2, but use trade and capital account liberalisation cutoffs ranging from the 40th (p40) to the 60th (p60) percentiles. Statistical significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. For all other details see Table 2.

account liberalisation 0.98 ($t = 2.5$) and 0.51 ($t = 2.5$). The modest attenuation is consistent with a small tax-transition component, but the predetermined capacity already delivers the gradient, so it does not merely reflect capacity responding to openness. Appendix D.6 extends this analysis to the baseline index, capacity measured in the 1980s and at the onset year.

Table 6: Alternative moderator variables: gradient per standard deviation.

	Income p.c.	Baseline capacity	Fiscal capacity
Trade liberalisation D_{tr}			
ETR _K , trade	+1.29** (0.55)	+2.05*** (0.78)	+2.04*** (0.70)
ETR _L , trade	+1.15*** (0.33)	+1.35*** (0.33)	+1.49*** (0.33)
Capital account liberalisation D_{fin}			
ETR _K , capital	+1.48** (0.71)	+1.97*** (0.69)	+1.93*** (0.68)
ETR _L , capital	+0.81*** (0.30)	+1.03*** (0.36)	+1.13*** (0.37)

Notes: Standard errors in parentheses. Each entry is the gradient of the country CCE effect $\hat{\beta}_i$ on the indicated moderator (within-country average), standardised so the coefficient is the change in the treatment effect per one standard deviation of the moderator (robust regression on the standardised moderator; $|t|$ in parentheses). State capacity and Fiscal capacity are taken from O'Reilly and Murphy (2022); income is average log GDP per capita. Statistical significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Alternative moderators That the gradient is about *state capacity* rather than income is supported by Table 6. Standardising each candidate moderator, the per-standard-deviation gradient is larger and more precisely estimated for both the baseline state-capacity index and the fiscal sub-index than for income per capita, in every specification.¹² Income is a coarser proxy for enforcement capacity, and the direct measures predict the treatment heterogeneity better. The two are strongly correlated, so this is a comparison of rival proxies rather than a decisive test between them; Appendix D.1 traces the gradient against income directly and finds the same sign change.

We focus our main results on the baseline index, which cannot reflect a tax-on-tax mechanism, as it excludes any fiscal or tax indicator. We report the fiscal sub-index, the most direct measure of the administrative-tax channel, as a robustness check that yields effectively the same gradient.¹³

The theoretical framework ties the *factor-share* response to capital abundance

¹²Using state capacity, treatment effects per standard-deviation are 59% (ETR_K) and 17% (ETR_L) larger than those for income per capita in the trade liberalisation case. For capital-account liberalisation these figures are 33% and 27%, respectively.

¹³A placebo confirms the gradient is not mechanical: randomly reassigning the capacity index across countries (2,000 permutations) yields $\hat{\gamma}_1$ centred on zero (mean ≈ 0 , s.d. $\approx$ the analytic standard error), and the actual gradient lies far in the tail ($p \leq 0.009$ for all four outcomes).

Table 7: Separating the channels: factor shares track abundance, taxes track capacity.

	Trade liberalisation (D_{tr})	
	State capacity (A2)	Capital abundance (A1)
Capital share θ_K	−1.18* (0.68)	+1.02* (0.61)
Labour share θ_L	+1.18* (0.68)	−1.02* (0.61)
Capital ETR	+2.31** (0.91)	−0.35 (0.60)
Labour ETR	+1.06*** (0.40)	+0.41 (0.36)

Notes: Standard errors in parentheses. Each row regresses the country-specific trade-liberalisation treatment effect $\hat{\beta}_i$ on *both* standardised moderators jointly by robust (Huber) regression; the two columns report the two coefficients, the change in the effect per standard deviation of each moderator, holding the other constant, with $|t|$ reported in parentheses. State capacity is the [O'Reilly and Murphy \(2022\)](#) index used in the baseline results; capital abundance is the country-mean log capital–labour ratio $\ln(K/L)$ from the Penn World Table. $N_{tr} = 55$ trade liberalisers. Statistical significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

(Assumption A1, Stolper–Samuelson) but the *effective-rate* response to administrative capacity (A2); a single capacity moderator cannot tell the two apart. We therefore give each channel its own correlate. Alongside the baseline state-capacity index we add capital abundance, the country's mean log capital–labour ratio $\omega_i = \ln(K_i/L_i)$ from the Penn World Table, and regress each per-country trade effect on both (standardised). Because the capacity index excludes any tax indicator, the capacity channel cannot be a tax-on-tax artefact. The channels separate (Table 7; capacity and abundance are themselves correlated at 0.69 across these countries, so each coefficient is netted of the other). The capital- and labour-share responses, though imprecisely estimated, load on *abundance* with the equal-and-opposite sign Stolper–Samuelson predicts (+1.02 and −1.02 per standard deviation, each significant at the 10% level). The capital effective-rate response, by contrast, moves with *state capacity* (2.31, $t = 2.5$), not with abundance once capacity is held constant (−0.35, $t = 0.6$). The labour effective rate behaves the same way (1.06, $t = 2.7$, against 0.41, $t = 1.1$). The development-graded effect on *taxation* is thus an administrative-capacity phenomenon rather than a factor-endowment one (A2, not A1, is the operative channel for taxes). The share gradient tracks abundance, as Stolper–Samuelson predicts, though, as in the main results, the share estimates are too imprecise to pin down a magnitude: only their sign is informative.

Two features of the share rows look odd at first sight and are worth flagging. The two rows are mirror images by construction: within columns, the capital and labour shares sum to one, so any moderator's coefficients on θ_K and θ_L are equal and opposite and carry identical t -ratios. And within each row the capacity and abundance coefficients take opposite signs (−1.18 against +1.02 for θ_K). This is what partialling out a collinear regressor produces, not a second channel: the two moderators correlate at 0.69, the share response is an abundance phenomenon, and the capacity coefficient is the remainder once abundance is held fixed. It is

significant at the 10% level at best, and we do not read it as capacity depressing the capital share. The content of the table is the contrast between the two pairs of rows: shares track abundance, effective rates track capacity.

An alternative labour ETR The labour effective rate of [Bachas et al. \(2023\)](#) includes social security contributions (SSC), which in these economies are large: about half of all labour taxes in high-income countries, a share that itself rises with state capacity (from 0.31 in developing countries to 0.47 in high-income ones). Because contributions finance pay-as-you-go pensions and health insurance, they rise with population ageing, so part of the labour effective rate, and of its capacity gradient, could reflect demographics and the size of the welfare state rather than the capacity to tax labour income.

We therefore strip social security contributions out of the labour tax and re-estimate the baseline gradient on an income-tax-only labour rate, $ETR_L^{\text{inc}} = (\tau_L - \text{SSC})/\theta_L$, using the same estimator and state capacity points as Table 2. Table 8 reports the result with and without contributions; the 'ETR_L' columns reproduce the baseline. Excluding contributions cuts the gradient by roughly half, from 0.94 to 0.39 for trade liberalisation and from 0.70 to 0.42 for capital-account liberalisation, but it stays positive and highly significant on both, and the predicted effect at high capacity remains so, at +1.05 and +1.07 percentage points (down from +2.71 and +1.68). The labour result is thus not an artefact of social-insurance financing or demographics: the capacity to tax labour *income* rises with state capacity under liberalisation, as it does for capital. The capital effective rate is unaffected, since contributions fall on labour alone.

5.4 Do the liberalisation episodes capture real integration?

Because our treatment is a *de jure* switch, Table 9 checks that the liberalisation episodes move *de facto* volumes, for trade and capital alike. We confirm that each reform moves its own outcome: trade liberalisation raises the trade-to-GDP ratio by about 10%, and capital-account liberalisation raises the stock of foreign direct investment by a quarter to a third. More striking is what each reform does to the *other* channel's volume: trade liberalisation raises FDI by more than 40%, larger than the own effect of capital-account liberalisation, while capital-account liberalisation in turn modestly but significantly lifts trade (by a similar margin as trade liberalisation). The two reforms move FDI together because the same firms drive both: multinationals account for the bulk of trade as well as direct investment, so opening either channel pulls in direct investment. Total external assets and liabilities (IFI), dominated by debt and valuation swings, show no systematic response to either reform.

This bears on the [Bachas et al. \(2023\)](#) interpretation: because trade liberalisation raises FDI so strongly, *de facto* trade openness moves closely with financial integration. The correlation between trade openness and capital taxation is therefore, in large part, a mechanical

response of the mobile capital that liberalisation brings in, not of trade as such.¹⁴

The second and fourth columns of Table 9 show how each of these responses varies with capacity, and the pattern is not uniform. Trade opening raises the trade-to-GDP ratio by about the same proportion everywhere: its capacity gradient is flat (+0.04 per standard deviation of capacity, $t = 0.9$). Its effect on FDI is both larger and capacity-graded (+0.45 on average, gradient +0.17, $t = 2.3$). Capital-account opening is graded on both counts, lifting trade by +0.08 with a gradient of +0.11 ($t = 3.3$) and FDI by +0.30 with a gradient of +0.14 ($t = 2.3$). These gradients and the mean effects are directly comparable: one standard deviation adds 0.17 to a mean FDI response of 0.45, about two-fifths of its size. The inward FDI stock, the part of that position a host state actually taxes, does not follow this pattern: it rises substantially with liberalisation (+0.44 and +0.25) but shows no significant capacity gradient (+0.12 and +0.06; $t = 1.4$ and 0.9).

Table 8: The labour gradient and social security contributions.

	Trade liberalisation		Capital account lib.	
	(1) ETR _L	(2) excl. SSC	(3) ETR _L	(4) excl. SSC
SC Gradient $\hat{\gamma}_1$	+0.94*** (0.229)	+0.39*** (0.104)	+0.70*** (0.244)	+0.42*** (0.124)
$\hat{\beta}$ (low SC)	-0.34 (0.41)	-0.21 (0.15)	-0.59 (0.46)	-0.29 (0.18)
$\hat{\beta}$ (medium SC)	+0.93*** (0.34)	+0.31** (0.15)	+0.35 (0.31)	+0.27 (0.17)
$\hat{\beta}$ (high SC)	+2.71*** (0.61)	+1.05*** (0.30)	+1.68*** (0.55)	+1.07*** (0.34)
N_{tr}	55	55	50	50

Notes: As in Table 2: the Huber M-estimator gradient $\hat{\gamma}_1$ of the country-specific liberalisation effect on the baseline state-capacity index, and the predicted effect $\hat{\beta}$ at the low, medium and high capacity-tercile centroids of state capacity (SC). The ETR_L columns reproduce the baseline labour effective rate results (with social security included); ‘excl. SSC’ uses the income-tax-only rate $ETR_L^{inc} = (\tau_L - SSC)/\theta_L$, removing social security contributions (OECD category 2000, which fall entirely on labour). N_{tr} is the number of liberalising countries. Statistical significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. For all other details see Table 2.

How the tax result reads turns on whether it reflects the ability to tax an opening or simply having more of it to tax, and the weight falls on the former. Our baseline outcomes are effective

¹⁴Estimating each liberalisation, on all liberalisers, against a single control pool of countries that never liberalised on either dimension (Appendix B, Table B-1) confirms the gradient: both trade and capital-account liberalisation raise the capital and labour effective rates in high-capacity states. We estimate each onset against this common pool rather than entering both in one regression because the two onsets often coincide (median gap 3 years; 21 of the 47 countries that liberalised on both did so within two years).

Table 9: *De jure* liberalisation raises *de facto* openness, for trade and capital alike.

Outcome (in logs)	Trade liberalisation		Capital liberalisation	
	(1) MG effect	(2) Capacity gradient	(3) MG effect	(4) Capacity gradient
Trade openness	+0.10** (0.05)	+0.04 (0.05)	+0.08* (0.04)	+0.11*** (0.03)
FDI openness	+0.45*** (0.08)	+0.17** (0.07)	+0.30*** (0.07)	+0.14* (0.06)
Inward FDI stocks	+0.44*** (0.08)	+0.12 (0.09)	+0.25*** (0.07)	+0.06 (0.07)
Total IFI	−0.01 (0.05)	+0.05 (0.05)	+0.01 (0.06)	+0.07 (0.08)

Notes: Standard errors in parentheses. Each cell is the mean-group average of the country-specific CCE effect of the liberalisation onset, trade or capital-account, on the indicated measure of *de facto* openness, all in logs, with *t*-statistic in parentheses below; the ‘capacity gradient’ columns give the robust slope of that country effect on baseline state capacity, per standard deviation. Trade openness is $(X+M)/GDP$ from the Penn World Table; the capital measures are from the External Wealth of Nations database (Lane and Milesi-Ferretti 2018): total international financial integration (foreign assets + liabilities)/GDP, FDI openness (FDI assets + liabilities)/GDP, and FDI stocks (inward FDI liabilities)/GDP. The number of liberalising countries is 55 for trade liberalisation and 49 to 50 for capital-account liberalisation. Statistical significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

tax rates, revenue over the matching factor income, so a larger base raises numerator and denominator alike and cannot mechanically lift the rate: the rate gradient is by construction about how much is collected per unit of base, not the size of the base. The base evidence points the same way: the inward stock the state actually taxes carries no significant gradient, so a modest exposure channel cannot be excluded but the base gives no clear sign of favouring capable states. So the gradient is better read as capable states taxing a common opening more effectively rather than attracting more of it to tax.

6 Discussion

Our central finding, that the effect of liberalisation on factor taxation is graded by state capacity, bears directly on the recent reassessment by Bachas et al. (2023). We begin with their reassessment, then relate our results to four strands of the literature.

Country-level evidence Bachas et al. (2023) set a divide between developing and rich countries; letting the effect vary with a country’s state capacity, we read a continuous gradient in the capacity to tax. The facts behind their framing are not in dispute: capital tax rates rose in poor countries and fell in rich ones. Where we differ from their conclusion is on the causes. Globalisation is not the main driver: much of the rise is a secular trend shared by liberalising and non-liberalising countries alike (Figure 2; the insignificant ATET estimates in

Table 2). Where liberalisation does bite, it is graded by state capacity, which income proxies only coarsely: it erodes the tax base of the poorest and lifts revenue only where the state can already tax. The gains accrue to richer economies, not to the developing world at large.

One way to reconcile our results with theirs is that [Bachas et al. \(2023\)](#) identify a *local* high-capacity effect in a handful of large emerging economies that attract foreign capital. Their mechanism raises taxation only where the state can reach the formal corporate sector, and in such countries that reach is concentrated in export zones and special economic zones that tax far more effectively than the country at large. China is the clearest case: its capital effective tax rate rose from 5% to 25% as activity moved into the coastal zones, though by the statutory measure it never liberalised and sits in our control group. The poorest liberalisers lie outside that reach, and they are where we estimate erosion. Independent revenue data corroborate both readings: re-estimating the design on the ICTD/UNU-WIDER Government Revenue Dataset reproduces the capacity-graded rise in income and corporate tax, and the same data show China as a high-mobilisation tax state, non-resource tax near 24% of GDP, despite a flat V-Dem capacity score (Appendix C).

The country-level evidence supports the state capacity finding directly. Figure 4 plots, for developing and upper-middle economies, the change in the capital effective rate against the change in state capacity between the pre- and post-1995 periods: the two rise together (correlation +0.27). Rwanda (used for illustration in [Bachas et al. 2023](#)) is the cleanest vignette: panel (b) traces its baseline capacity and capital effective rate over 1980–2018, the rate roughly doubling from about 5 to 11% as capacity rose; the milder post-minus-pre-1995 mean change is what enters panel (a) as its labelled point.

Relationship to the Literature We now relate our findings to four strands of the literature. First, the capacity-graded effect of liberalisation on capital tax revenue and the capital effective rate points to an *enforcement-capacity channel* (Proposition 2). The predicted τ_K response runs from a contraction in low-capacity states to an expansion of roughly one percentage point of net domestic product in high-capacity ones. The expansion in richer economies is consistent with stronger tax administration capturing the capital income that openness mobilises; the contraction in low-capacity countries is consistent with profit-shifting evidence ([Tørsløv et al. 2023](#)), whereby capital openness under limited administrative capacity enables base erosion rather than revenue capture. This low-capacity erosion is the effective-rate counterpart of the fiscal cost of trade liberalisation documented by [Cagé and Gadenne \(2018\)](#): across two centuries of revenue data, countries that liberalised *before* building fiscal capacity failed to replace lost tariff revenue with domestic taxes, a cost borne almost entirely by low-capacity developing economies. Our high-capacity expansion is its mirror image, the successful tax transition in which a capable state shifts onto domestic factor taxation, so the capacity gradient places both outcomes on a single dimension: the fiscal capacity that [Cagé and Gadenne \(2018\)](#) invoke to explain the cost but, like [Baunsgaard and Keen \(2010\)](#) and [Bachas et al. \(2023\)](#), proxy with the level of development.

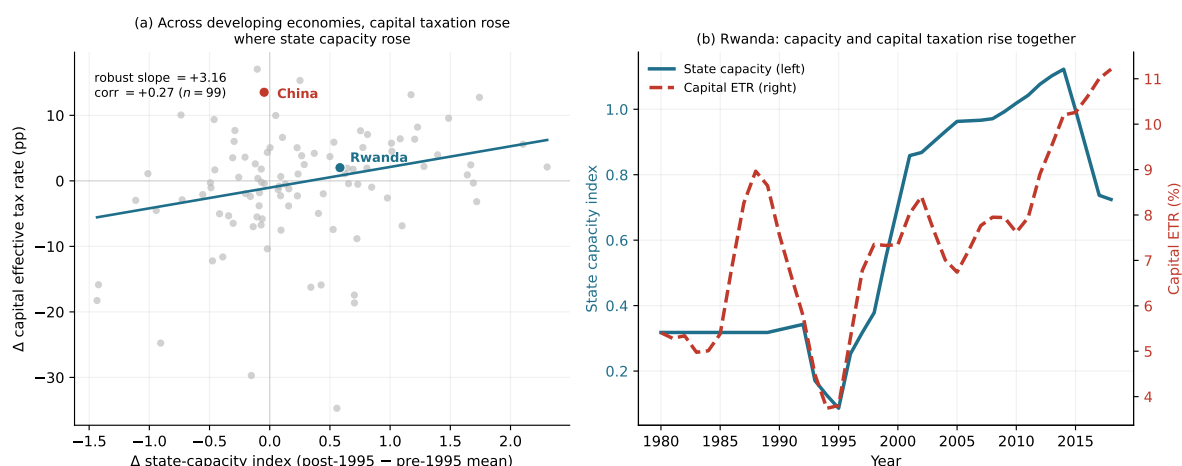


Figure 4: Capital taxation rose where state capacity rose. (a) Each point is a developing or upper-middle-income economy: the change in its capital effective rate against the change in its baseline state-capacity index (post-1995 minus pre-1995 means); the two are positively related (the line is a robust Huber regression; its slope and the correlation are shown). Only the two cases discussed in the text are labelled, Rwanda (the vignette of panel b) and China, the telling exception whose *national* capacity score is flat yet whose capital rate rose steeply as capacity was built *locally* in the coastal zones; all other economies are left unlabelled. (b) Rwanda: state capacity and the capital effective rate, 1980–2018.

Second, the labour effective rate rises with capacity alongside the capital rate, and a country's two responses tend to line up: across liberalisers, a country's estimated effect on its *capital* effective rate and its effect on its *labour* effective rate are positively correlated (+0.5 under trade liberalisation), so for the most part the states where opening raised capital taxation are those where it raised labour taxation, and the states where it eroded one are those where it eroded the other. A pure shift of the burden from mobile capital onto labour would move them in opposite directions; that they instead move together, on balance, points to a common driver, the state's capacity to tax, which captures more of *both* factors' income where enforcement is strong (Proposition 3). Globalisation thus forces no systematic capital-versus-labour trade-off, the shift of the tax burden onto less-mobile factors that Rodrik (1997) argued and that Egger, Nigai, and Strecker (2019) document for high-income economies, where globalisation raised the labour-tax burden of the less-mobile middle class: where the state is capable, opening up lifts the effective taxation of both; where it is not, both stay flat or fall. That the gradient reappears in labour, a separate tax base, also guards against a capital-specific measurement explanation for our findings. On the direction of the labour effect we align with Bachas et al. (2023), who likewise find trade integration raising labour taxation in high-income economies.

Third, the factor-reallocation channel of trade theory appears in our estimates only as a directional regularity (Stolper and Samuelson 1941, Karabarbounis and Neiman 2014). The capital- and labour-share gradients are equal and opposite by construction, since the two shares sum to one, and neither is individually significant: the per-country share responses are too imprecisely estimated to identify the gradient. The contrast with the precisely-

estimated tax gradients is the paper’s central substantive message. The development-graded effect of globalisation on effective taxation operates through the *tax-policy and enforcement* components, the statutory rate κ_f and the captured share ϕ_f , rather than through factor reallocation θ_f .

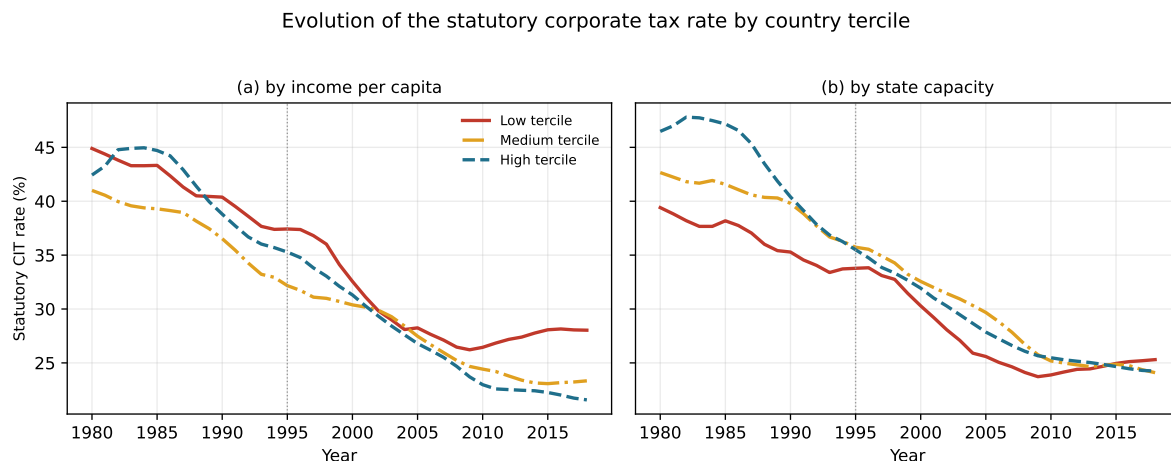


Figure 5: Statutory versus effective capital taxation: the race to the bottom is in the headline rate. The statutory corporate income tax rate ([Tax Foundation 2025](#)) for low, medium, and high tercile countries, sorted by income per capita (a) and state capacity (b), 1980–2018 (three-year moving average). Read against the effective capital rate in Figure 1: statutory rates converge across capacity, as the high tercile cuts its headline rate from about 46 to 25%, yet effective rates stay stratified, so the effective-to-statutory ratio ETR/κ rises with capacity, as the captured-share channel ϕ predicts.

Fourth, our results refine rather than overturn the canonical tax-competition prediction that capital mobility erodes capital taxation ([Zodrow and Mieszkowski 1986](#), [Wilson 1986](#), [Devereux, Lockwood, and Redoano 2008](#)). Three observations reconcile the two. First, tax competition is fought over *statutory* rates, whereas we measure *effective* rates and revenue: governments may cut headline rates to attract mobile capital even as the effective rate rises, because openness broadens the captured base, the mechanism [Bachas et al. \(2023\)](#) document for the developing-country corporate sector. We confirm this split directly in Figure 5: Statutory corporate rates ([Tax Foundation 2025](#)) fell across the board, the canonical race to the bottom, and in our estimates trade liberalisation lowers them by about two percentage points. But they fell furthest where state capacity, and the starting rate, were highest: the top-capacity tercile cut its headline rate from roughly 46 to 25%, converging onto the bottom tercile. Effective capital rates, by contrast, did not converge across countries (Figure 1): the high-capacity gap persisted. A converging statutory rate alongside a stratified effective rate means the ratio of effective to statutory taxation, ETR/κ , rises with capacity.¹⁵ The headline rate, over which tax competition is fought, falls, while what high-capacity states actually

¹⁵Because ETR_K taxes all capital income while κ is the corporate statutory rate, this ratio is a collection proxy and can exceed unity; it is consistent with, but not a direct measurement of, the bounded captured share ϕ of the model.

collect holds up because openness widens the captured base, as the captured-share channel of $ETR = \kappa\phi$ predicts. Second, the base erosion that tax competition predicts does appear in our estimates, but at the *low-capacity* end of the distribution, where the predicted effect of capital opening on the capital tax base is negative, consistent with profit shifting and capital flight under weak enforcement (Tørsløv et al. 2023). Third, the tax-competition prediction that the burden shifts onto the immobile factor is borne out in the statutory rates, where the corporate rate falls; the effective labour rate rises alongside the capital rate, through the same enforcement. The capacity gradient we estimate is thus the net of two forces the competition literature itself emphasises: *base erosion through mobility*, which dominates where enforcement capacity is weak, and *base capture through enforcement*, which dominates where it is strong.

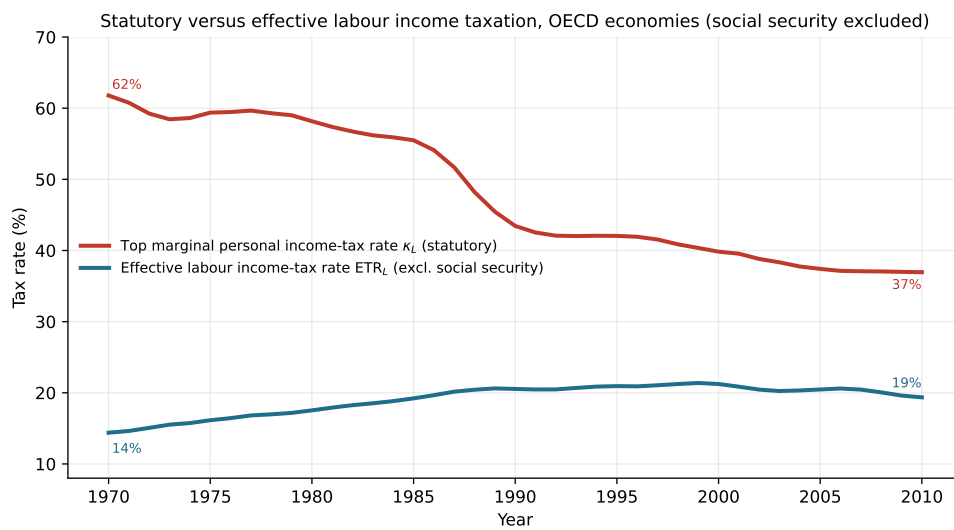


Figure 6: The labour counterpart of Figure 5, on income taxation, for the OECD economies with long series. The top marginal personal income tax rate (κ_L ; Comparative Income Taxation Database, Genovese et al. 2016) and the effective labour income-tax rate (ETR_L , excluding social security contributions), cross-country averages, 1970–2010 (three-year moving average). The statutory rate falls while the effective rate rises, so the collected share of the statutory rate rises, as for capital. Contributions are excluded because they are pay-as-you-go and demographically driven rather than an enforcement margin (Section 5.3). Illustrative only: these are high-income economies, with no comparable series for lower-capacity countries.

The same pattern holds on the labour side, for income taxation (see Figure 6). In the rich economies for which long top-rate series exist, the top marginal personal income tax rate fell over the liberalisation era, from about 62% in the 1970s to 38% in the 2000s, yet the effective labour income-tax rate rose, from about 16% to 20%. We exclude social security contributions here, so that both series are income taxes and the comparison is not driven by demographics. As for capital, the statutory rate fell while the effective rate rose, so the share of the headline rate actually collected roughly doubled, an enforcement phenomenon, not a statutory one.¹⁶

¹⁶ κ_L is a top marginal rate and ETR_L an average effective rate, so their ratio is a collection proxy rather

7 Concluding Remarks

Whether globalisation helps or hinders the taxation of capital depends on the state's capacity to tax. Studying *de jure* liberalisation and letting each country respond to opening up in its own way, we find its effect on factor taxation graded by capacity: it raises the effective taxation of both capital and labour where the state is strong and erodes both where it is weak. The two tend to rise together, their responses positively correlated rather than offsetting, so opening up forces no systematic trade-off between taxing capital and labour: where the state can tax, the same enforcement lifts both.

This conclusion departs from the reassessment that motivated our paper, though it rests on the same facts. We do not dispute what [Bachas et al. \(2023\)](#) document, that capital taxation *rose* in developing countries and *fell* in rich ones; we argue for a different cause. Globalisation is not the main driver of the developing-country rise, much of which is a broader trend, and the part it does drive reaches the already-capable countries, not the poorest. Nor is the rich-country fall a consequence of greater openness, since liberalisation raises the effective rate where state capacity is high.

The implication for policy is that the fiscal consequences of opening up to the world economy are not uniform across the development spectrum. The same liberalisation that broadens the tax base and lifts revenue where tax administration is strong can erode it where administration is weak. Building the capacity to assess and collect taxes is a *precondition* for openness to yield fiscal gains rather than losses, a sequencing lesson for developing economies and the institutions that advise them. Our analysis also reconciles the long-standing fear that mobile capital forces a race to the bottom with the revenue gains that open, capable states in fact enjoy: capital mobility erodes the tax base only where the state lacks the means to defend it.

References

- Ahn, S. C. and A. R. Horenstein (2013). "Eigenvalue Ratio Test for the Number of Factors." *Econometrica* 81(3), 1203–1227.
- Bachas, P., M. H. Fisher-Post, A. Jensen, and G. Zucman (2023). "Globalization, Capital Taxation and Development: Evidence from a Macro-Historical Database." Forthcoming, *American Economic Journal: Applied Economics* (NBER Working Paper 29819).
- Bachas, P., L. Gadenne, and A. Jensen (2024). "Informality, Consumption Taxes and Redistribution." *Review of Economic Studies* 91(5), 2604–2634.
- Baunsgaard, T. and M. Keen (2010). "Tax Revenue and (or?) Trade Liberalization." *Journal of Public Economics* 94(9–10), 563–577.

than the bounded ϕ_L of the model.

- Bai, J. (2009). "Panel Data Models with Interactive Fixed Effects." *Econometrica* 77(4), 1229–1279.
- Borusyak, K., X. Jaravel, and J. Spiess (2024). "Revisiting Event-Study Designs: Robust and Efficient Estimation." *Review of Economic Studies* 91(6): 3253–3285.
- Cagé, J. and L. Gadenne (2018). "Tax Revenues and the Fiscal Cost of Trade Liberalization, 1792–2006." *Explorations in Economic History* 70, 1–24.
- Callaway, B. and P. H. C. Sant'Anna (2021). "Difference-in-Differences with Multiple Time Periods." *Journal of Econometrics* 225(2): 200–230.
- Chan, M. K. and S. S. Kwok (2022). "The PC-DID Approach: Difference-in-Differences when Trends are Potentially Unparallel and Stochastic." *Journal of Business & Economic Statistics* 40(3): 1216–1233.
- Chinn, M. D. and H. Ito (2006). "What Matters for Financial Development? Capital Controls, Institutions, and Interactions." *Journal of Development Economics* 81(1): 163–192.
- Chudik, A. and M. H. Pesaran (2015). "Common Correlated Effects Estimation of Heterogeneous Dynamic Panel Data Models with Weakly Exogenous Regressors." *Journal of Econometrics* 188(2): 393–420.
- de Chaisemartin, C. and X. D'Haultfœuille (2020). "Two-Way Fixed Effects Estimators with Heterogeneous Treatment Effects." *American Economic Review* 110(9), 2964–2996.
- De Visscher, S., Eberhardt, M., and Everaert, G. (2020). "Estimating and testing the multicountry endogenous growth model." *Journal of International Economics*, 125, 103325.
- Devereux, M. P., B. Lockwood, and M. Redoano (2008). "Do Countries Compete Over Corporate Tax Rates?" *Journal of Public Economics* 92(5–6): 1210–1235.
- Eberhardt, M. (2022). "Democracy, growth, heterogeneity, and robustness." *European Economic Review*, 147, 104173.
- Egger, P. H., S. Nigai, and N. M. Strecker (2019). "The Taxing Deed of Globalization." *American Economic Review* 109(2): 353–390.
- Eberhardt, M., Helmers, C., and Strauss, H. (2013). "Do spillovers matter when estimating private returns to R&D?" *Review of Economics and Statistics*, 95(2), 436–448.
- Eberhardt, M., and Presbitero, A. F. (2015). "Public debt and growth: Heterogeneity and non-linearity." *Journal of International Economics*, 97(1), 45–58.
- Eberhardt, M., and Teal, F. (2020). "The magnitude of the task ahead: Macro implications of heterogeneous technology." *Review of Income and Wealth*, 66(2), 334–360.

- Ertur, C., and Musolesi, A. (2017). "Weak and strong cross-sectional dependence: A panel data analysis of international technology diffusion." *Journal of Applied Econometrics*, 32(3), 477-503.
- Estefania-Flores, J., D. Furceri, S. Hannan, J. D. Ostry, and A. K. Rose (2022). "A Measurement of Aggregate Trade Restrictions and Their Economic Effects." *IMF Working Paper* 22/1.
- Feenstra, R. C., R. Inklaar, and M. P. Timmer (2015). "The Next Generation of the Penn World Table." *American Economic Review* 105(10): 3150–3182.
- Genovese, F., K. Scheve, and D. Stasavage (2016). *Comparative Income Taxation Database*. Stanford University, <http://data.stanford.edu/citd>.
- Gobillon, L. and T. Magnac (2016). "Regional Policy Evaluation: Interactive Fixed Effects and Synthetic Controls." *Review of Economics and Statistics* 98(3), 535–551.
- Goodman-Bacon, A. (2021). "Difference-in-Differences with Variation in Treatment Timing." *Journal of Econometrics* 225(2): 254–277.
- Hanson, J. K. and R. Sigman (2021). "Leviathan's Latent Dimensions: Measuring State Capacity for Comparative Political Research." *Journal of Politics* 83(4): 1495–1510.
- Jensen, A. D. and J. L. Weigel (2025). "No Taxation Without Administration: Bringing the State Back into the Public Finance of Developing Countries." *Journal of Economic Literature*. Forthcoming.
- Lewis, J. B. and D. A. Linzer (2005). "Estimating Regression Models in Which the Dependent Variable Is Based on Estimates." *Political Analysis* 13(4): 345–364.
- Karabarbounis, L. and B. Neiman (2014). "The Global Decline of the Labor Share." *Quarterly Journal of Economics* 129(1): 61–103.
- Lane, P. R. and G. M. Milesi-Ferretti (2018). "The External Wealth of Nations Revisited: International Financial Integration in the Aftermath of the Global Financial Crisis." *IMF Economic Review* 66(1), 189–222.
- O'Reilly, C. and R. H. Murphy (2022). "An Index Measuring State Capacity, 1789–2018." *Economica* 89(355), 713–745.
- Pesaran, M. H. (2004). "General Diagnostic Tests for Cross Section Dependence in Panels." *CESifo Working Paper* 1229.
- Phipson, B. and G. K. Smyth (2010). "Permutation *P*-values Should Never Be Zero: Calculating Exact *P*-values When Permutations Are Randomly Drawn." *Statistical Applications in Genetics and Molecular Biology* 9(1), Article 39.

- Pesaran, M. H. (2006). "Estimation and Inference in Large Heterogeneous Panels with a Multifactor Error Structure." *Econometrica* 74(4): 967–1012.
- Pesaran, M. H. and Y. Xie (2023). "A Bias-Corrected CD Test for Error Cross-Sectional Dependence in Panel Data Models with Latent Factors." *Working paper*.
- M. H. Pesaran and R. P. Smith (1995). "Estimating long-run relationships from dynamic heterogeneous panels." *Journal of Econometrics* 68(1): 79–113.
- Rodrik, D. (1997). *Has Globalization Gone Too Far?* Washington, DC: Institute for International Economics.
- Stolper, W. F. and P. A. Samuelson (1941). "Protection and Real Wages." *Review of Economic Studies* 9(1): 58–73.
- Sun, L. and S. Abraham (2021). "Estimating Dynamic Treatment Effects in Event Studies with Heterogeneous Treatment Effects." *Journal of Econometrics* 225(2): 175–199.
- Tax Foundation (2025). *Corporate Tax Rates around the World, 2025*. Available online: <https://taxfoundation.org/data/all/global/corporate-tax-rates-by-country-2025/>
- Tørsløv, T. R., L. S. Wier, and G. Zucman (2023). "The Missing Profits of Nations." *Review of Economic Studies* 90(3): 1499–1534.
- UNU-WIDER (2025). *Government Revenue Dataset*. International Centre for Tax and Development / UNU-WIDER. Available online: <https://www.wider.unu.edu/project/grd-government-revenue-dataset>.
- Wamser, G., V. Merlo, M. Ruf, et al. (2026). "The ITI Database: New Data on International Tax Institutions." *International Tax and Public Finance* 33: 734–762.
- Wilson, J. D. (1986). "A Theory of Interregional Tax Competition." *Journal of Urban Economics* 19(3): 296–315.
- Zodrow, G. R. and P. Mieszkowski (1986). "Pigou, Tiebout, Property Taxation, and the Underprovision of Local Public Goods." *Journal of Urban Economics* 19(3): 356–370.

Online Appendix: not intended for publication

A Country classification by liberalisation status

Table A-1 classifies the 147 countries by whether they liberalised trade, the capital account, both, neither, or were always open during the sample period.

Table A-1: Countries by liberalisation status.

Group	<i>N</i>	Countries (ISO3)
Both	47	ARG, AUS, BGR, BOL, BRA, BWA, CHL, CRI, DNK, DOM, ECU, EGY, ESP, FIN, FRA, GBR, GMB, GRC, GTM, HND, HRV, HUN, IRL, ISR, ITA, JAM, KEN, KHM, KOR, LKA, MEX, NIC, NOR, NZL, PER, PHL, PRT, PRY, ROU, RUS, RWA, SDN, SLV, TUR, UGA, VEN, ZMB
Trade only	13	ALB, BDI, BEL, COL, GHA, IDN, KAZ, MDG, MLI, SLE, THA, ZAF, ZWE
Capital only	14	AZE, CZE, GEO, HTI, JPN, KGZ, MNG, MYS, PNG, POL, SGP, SVK, SVN, YEM
Never liberalised (either) [‡]	35	BEN, BFA, BGD, BLR, BRB, CAF, CHN, CIV, CMR, COG, ETH, FJI, GAB, GIN, GNQ, IND, LAO, LSO, MAR, MMR, MOZ, MRT, NAM, NGA, NPL, PAK, SEN, SWZ, SYR, TGO, TUN, UKR, UZB, VNM, WSM
Always open (excluded)	38	ARM, AUT, BHR, BHS, BIH, BLZ, CAN, CHE, CYP, DEU, DZA, EST, GUY, ISL, JOR, KWT, LBN, LBR, LTU, LVA, MDA, MKD, MUS, MWI, NER, NLD, OMN, PAN, QAT, SAU, SLB, SWE, SYC, TCD, TTO, TZA, URY, USA
<i>Margin-specific control pools used in the main results (these overlap the groups above)</i>		
Trade controls	38	the 35 “Never liberalised” above, plus BLZ, MYS, NER
Capital-account controls	50	the 35 “Never liberalised” above, plus ALB, BDI, DZA, GHA, KAZ, MDA, MDG, MLI, MWI, SLB, TCD, THA, TZA, ZAF, ZWE

Notes: A country is a trade (capital-account) liberaliser if it has a clean liberalisation onset on that dimension during the sample; “Both” liberalised on both. “Always open (excluded)” collects the 38 countries that are open throughout on at least one dimension (or revert, or switch only briefly) and so enter no clean difference-in-differences sample *as liberalisers*; eight of them are never open on the other margin and serve there as controls (BLZ and NER on trade; DZA, MDA, MWI, SLB, TCD and TZA on the capital account). The five groups sum to the full 147. Country codes are ISO3.

The table classifies all 147 countries by *de jure* status alone, so its totals exceed the estimation samples of Section 3 due to sample cleaning. Of the 60 trade liberalisers (Both + Trade only), 55 enter the trade sample; the five dropped (ALB, HRV, KAZ, KHM, RUS) fail the 25-observation minimum. Of the 61 capital-account liberalisers (Both + Capital only), 50 enter the capital sample; the eleven dropped (AZE, CZE, GEO, HRV, KGZ, KHM, MNG, RUS, SVK, SVN, YEM) fail the same rule. These are thirteen distinct countries, HRV, KHM and RUS failing on both margins, all transition economies whose effective-rate series begin in the mid-1990s. The control pools are margin-specific and are listed at the foot of the table: by the statutory measure 38 countries are never open on trade and 50 never open on the capital account, of which 31 and 40 respectively clear the 25-observation minimum and enter the estimation. [‡] ‘Never liberalised’ marks the common control group (neither Trade nor Capital) used only for the joint estimation in Appendix B. In Appendix Section D.7 we revert some of the cleaning rules to find qualitatively identical results.

B Joint estimation against a common control pool

Table B-1 reports the effect of each liberalisation, on all liberalisers, against the control pool of countries that never liberalised *on either dimension*; it is discussed in Section 5.4.

Table B-1: Treatment effects against a common never-liberalised control pool.

	SC gradient $\hat{\gamma}_1$	$\hat{\beta}$ (low)	$\hat{\beta}$ (medium)	$\hat{\beta}$ (high)
ETR_K				
Trade liberalisation	+1.55*** (0.53)	−1.95** (0.81)	+0.15 (0.68)	+3.11** (1.39)
Capital-account liberalisation	+1.32*** (0.43)	−1.80* (1.02)	−0.02 (0.73)	+2.49** (0.97)
ETR_L				
Trade liberalisation	+0.82*** (0.23)	−0.38 (0.39)	+0.73** (0.33)	+2.30*** (0.62)
Capital-account liberalisation	+0.69*** (0.23)	−0.68 (0.47)	+0.25 (0.31)	+1.57*** (0.50)

Notes: Per-country CCE estimates of each liberalisation, on all liberalisers, against a common control pool of countries that never liberalised *on either dimension*. $\hat{\gamma}_1$ is the robust slope of the country treatment effect on state capacity; $\hat{\beta}$ is the predicted effect at the bottom, middle, and top tercile of state capacity, as in Table 2. Statistical significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C Cross-source validation with the ICTD Government Revenue Dataset

Our effective tax rates are built from the [Bachas et al. \(2023\)](#) national-accounts data, so the capacity gradient could in principle be an artefact of that construction, the allocation of revenue to capital and labour and the division by factor income. We re-examine it with the ICTD/UNU-WIDER *Government Revenue Dataset* (GRD, 2025 vintage; 198 countries, 1980–2023), a harmonised compilation of government revenue by source as a share of GDP ([UNU-WIDER 2025](#)). The GRD is not independent of [Bachas et al. \(2023\)](#), who build their revenue numerator partly from the OECD Revenue Statistics and the ICTD, but it constructs the outcome differently, as total revenue over GDP with no allocation to capital or labour. We use it for three checks: (i) that our factor-tax aggregate tracks the GRD direct taxes; (ii) that the capacity gradient reappears when this differently constructed revenue replaces the effective rates; and (iii) that the raw revenue data show the same countries actually building fiscal capacity, which is what our capacity measure is meant to stand for.

C.1 The effective-rate series tracks the GRD

Figure C-1 plots the [Bachas et al. \(2023\)](#) total factor tax ($\tau_K + \tau_L$, share of NDP) against GRD direct taxes (income + property + social contributions, share of GDP) across 4,481 overlapping country-years (151 countries). The two move together almost one-for-one: the cross-sectional correlation is $+0.96$, and the *within-country* (country-demeaned) correlation is $+0.80$, so the series tracks this separately built source not only in levels but over time. The level gap reflects the broader factor attribution and NDP- versus GDP-scaling, not disagreement about movements. Since the two series share much of their underlying revenue data, this close tracking is a consistency check on our factor-tax aggregation, not an independent validation; the substantive check is the capacity gradient on the differently constructed GRD outcomes, below.

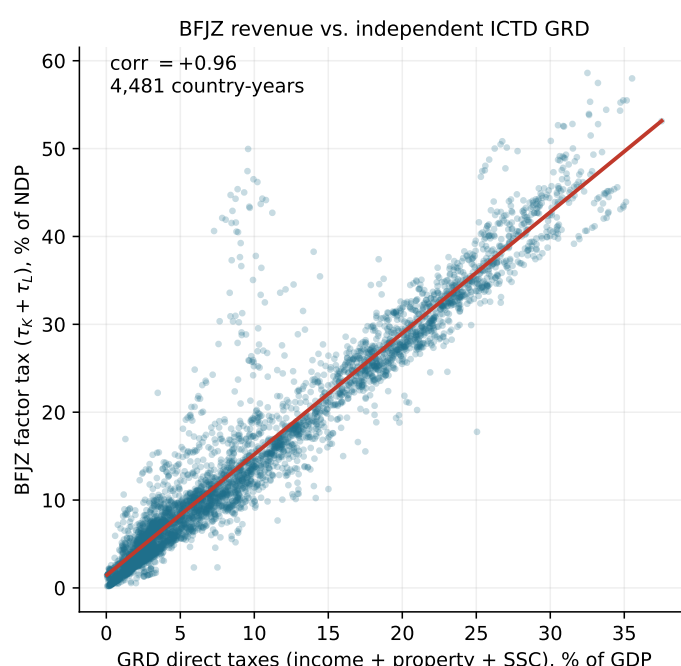


Figure C-1: Factor-tax revenue versus the ICTD/UNU-WIDER GRD direct taxes, all overlapping country-years.

C.2 The capacity gradient on GRD revenue

We re-run the baseline CCE-DID exactly as in the main text, per-country regressions of the outcome on the liberalisation onset and control cross-sectional averages, a robust (Huber-M) second-stage gradient on the baseline state-capacity index, and predicted effects at the capacity terciles, but replace the effective rates with GRD revenue measures (% of GDP) as the dependent variable. The outcome is a tax measure; the moderator remains the *non-tax* baseline state-capacity index, so there is no circularity. Table C-1 reports the gradient and the predicted effect at the low, medium and high capacity terciles.

Table C-1: Capacity gradient with GRD revenue as the dependent variable.

Tax component	(1) Income	(2) Corporate	(3) Trade	(4) Non-resource
Trade liberalisation				
SC Gradient $\hat{\gamma}_1$	+0.12 (0.10)	+0.09** (0.04)	−0.02 (0.04)	+0.09 (0.20)
$\hat{\beta}$ (low SC)	−0.02 (0.11)	−0.07 (0.08)	−0.07 (0.10)	+0.37 (0.46)
$\hat{\beta}$ (medium SC)	+0.15 (0.15)	+0.06 (0.08)	−0.10* (0.06)	+0.50 (0.36)
$\hat{\beta}$ (high SC)	+0.38 (0.31)	+0.23* (0.13)	−0.15** (0.06)	+0.67 (0.52)
N_{tr}	54	45	54	50
Capital-account liberalisation				
SC Gradient $\hat{\gamma}_1$	+0.26*** (0.10)	+0.15*** (0.06)	−0.06* (0.04)	+0.07 (0.16)
$\hat{\beta}$ (low SC)	−0.25 (0.16)	−0.16* (0.09)	−0.00 (0.09)	−0.17 (0.38)
$\hat{\beta}$ (medium SC)	+0.10 (0.15)	+0.05 (0.08)	−0.09 (0.06)	−0.07 (0.28)
$\hat{\beta}$ (high SC)	+0.59** (0.28)	+0.35** (0.15)	−0.21*** (0.06)	+0.06 (0.39)
N_{tr}	57	44	57	48

Notes: Per-country CCE-DID effect of the liberalisation onset on each GRD revenue measure (% of GDP), then the Huber-M gradient $\hat{\gamma}_1$ on the baseline state-capacity index, the predicted effect $\hat{\beta}$ at the low, medium and high capacity-tercile centroids; standard errors in parentheses. N is the number of liberalising countries with at least 20 GRD observations (a slightly looser threshold than the 25 used in the main text, reflecting the GRD's shorter coverage). Statistical significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The baseline pattern reappears in the raw revenue data. The *direct*-tax gradients, income and corporate tax, the GRD analogues of the capital effective rate, run from negative or zero at low capacity to positive at high, and are significant under capital-account liberalisation (income +0.26, corporate +0.15, both $p < 0.01$). Trade taxes *fall*, and fall more where capacity is high, as capable states replace tariff revenue with domestic taxation, the tax transition whose incompleteness in low-capacity countries [Cagé and Gadenne \(2018\)](#) document; total non-resource tax rises fairly uniformly, so the gradient is in the *composition* of revenue, not its level. Two caveats: GRD coverage is thinner than the effective-rate panel, so samples are smaller ($N = 44$ –57) and shift with the minimum-observation rule, and the magnitudes are not comparable to the effective-rate gradients, which are built differently. The *sign and shape*, negative at low capacity and positive at high, are the point, and they survive the switch to a separately constructed source.

C.3 The capacity gradient: direct vs consumption taxes

Liberalisation episodes of the 1980s and 1990s were often accompanied by value-added-tax adoption and tax-administration reform. If those reforms raised revenue, and did so more in capable states, they, rather than any change in the taxation of capital and labour, could generate the capacity gradient. We test this directly. The GRD records consumption taxes from 1980, so it covers the bulk of our onsets (median 1992 for trade, 1996 for capital); the [Wamser et al. \(2026\)](#) tax-institutions database, which records statutory VAT rates, begins only in 2001 and so post-dates three onsets in four, which is why we use the GRD here. Table C-2 re-runs the per-country CCE-DID with taxes on goods and services, their general (VAT-type) component, VAT proper, and total indirect taxes as the outcome. Consumption-tax revenue does rise at liberalisation, by a few tenths of a point of GDP, in line with the global spread of the VAT over these decades. But the rise carries *no* capacity gradient: every $\hat{\gamma}_1$ is small and statistically indistinguishable from zero (from -0.19 to $+0.11$), and the predicted low- and high-capacity effects coincide (not reported). Around the onset of liberalisation the series move smoothly rather than jumping: mean VAT revenue drifts by about a tenth of a point of GDP over the surrounding decade. The capacity grading is therefore specific to direct taxation, the income and corporate taxes that are the revenue analogue of the capital and labour effective rates, and is absent from the consumption taxes a coincident VAT reform would move. That direct taxes rather than consumption taxes carry the grading fits the broader observation that where administrative capacity is weak, redistribution runs through consumption taxes rather than direct ones ([Jensen and Weigel 2025](#), [Bachas, Gadenne, and Jensen 2024](#)). A consumption-tax reform that is uniform across capacity cannot manufacture a capacity-graded effective rate on capital and labour.

C.4 Revenue mobilisation in the raw data

Finally, the GRD lets us look at revenue mobilisation directly (Figure C-2). Panel (b) plots non-resource tax revenue in 2014–2023 against 1995–2004: 81% of developing economies lie above the 45-degree line, so these were decades of broad revenue mobilisation. Panel (a) tracks China’s revenue by source against the developing-economy average: its total non-resource tax approaches 24% of GDP, built on domestic consumption taxes, with tariff revenue driven to near zero. Neither panel plots state capacity, and GRD tax/GDP could not serve as our moderator in any case without re-introducing a tax-on-tax comparison. The inference we draw concerns our *measure*: China mobilises revenue like a high-capacity state while its V-Dem score stays flat throughout, so that score understates its fiscal-capacity building. China also illustrates the opposite of the fiscal cost [Cagé and Gadenne \(2018\)](#) document: it drove its own tariff revenue to near zero yet lifted total taxation to an OECD level on a domestic base, whereas the low-capacity liberalisers they study lose tariff revenue without replacing it.

Table C-2: CCE-DID with GRD consumption-tax revenue (% of GDP) as the outcome.

	Goods & services	General G&S (VAT-type)	VAT	Indirect tax
Trade liberalisation				
SC gradient $\hat{\gamma}_1$	-0.05 (0.17)	-0.01 (0.19)	-0.02 (0.22)	-0.19 (0.19)
Mean effect (MG) $\hat{\beta}$	+0.72*** (0.22)	+0.65*** (0.24)	+0.43* (0.24)	+0.39 (0.24)
N_{tr}	45	39	19	46
Capital-account liberalisation				
SC gradient $\hat{\gamma}_1$	-0.06 (0.15)	+0.08 (0.15)	+0.11 (0.13)	-0.08 (0.14)
Mean effect (MG) $\hat{\beta}$	+0.55** (0.22)	+0.20 (0.21)	+0.18 (0.16)	+0.06 (0.20)
N_{tr}	50	45	32	49

Notes: As Table C-1, with consumption-tax revenue as the dependent variable: taxes on goods and services, their general (VAT-type) component, VAT proper, and total indirect taxes (all % of GDP, ICTD/UNU-WIDER GRD). $\hat{\gamma}_1$ is the robust capacity gradient of the per-country onset effect; the mean effect is the robust (Huber-M) mean of those per-country effects. Standard errors in parentheses. N is the number of liberalising countries with at least 20 GRD observations. Statistical significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

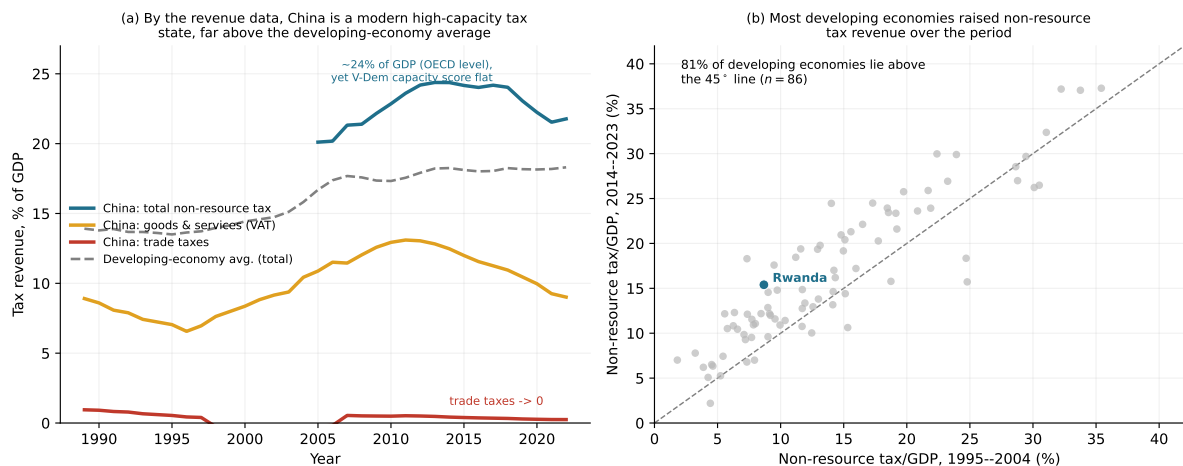


Figure C-2: Fiscal-capacity building in the ICTD GRD. (a) China's tax revenue by source (total non-resource tax, the goods-and-services/VAT component, and trade taxes), shown against the developing-economy average for total non-resource tax. China reaches an OECD-level $\approx 24\%$ of GDP on a broad domestic base, with tariffs near zero and far above its peers, although its V-Dem capacity score is flat. (b) Non-resource tax revenue, 1995–2004 versus 2014–2023, for developing economies; 81% lie above the 45° line. Rwanda is highlighted and all other economies are unlabelled; China does not appear in this panel because its GRD non-resource-tax series has no data in the early window.

D Robustness of the capacity gradient

D.1 The gradient against income per capita

Figure D-1 reproduces the running-line gradient of Figure 3 with average log GDP per capita in place of state capacity as the running variable, as a cross-check that the sign change is not specific to the capacity index. The same low-to-high pattern appears in every panel: the per-country effect on both effective rates rises from negative or zero at low income to positive at high income. Income is a coarser proxy for the enforcement capacity that drives the effect (Table 6), but the qualitative gradient is the same.

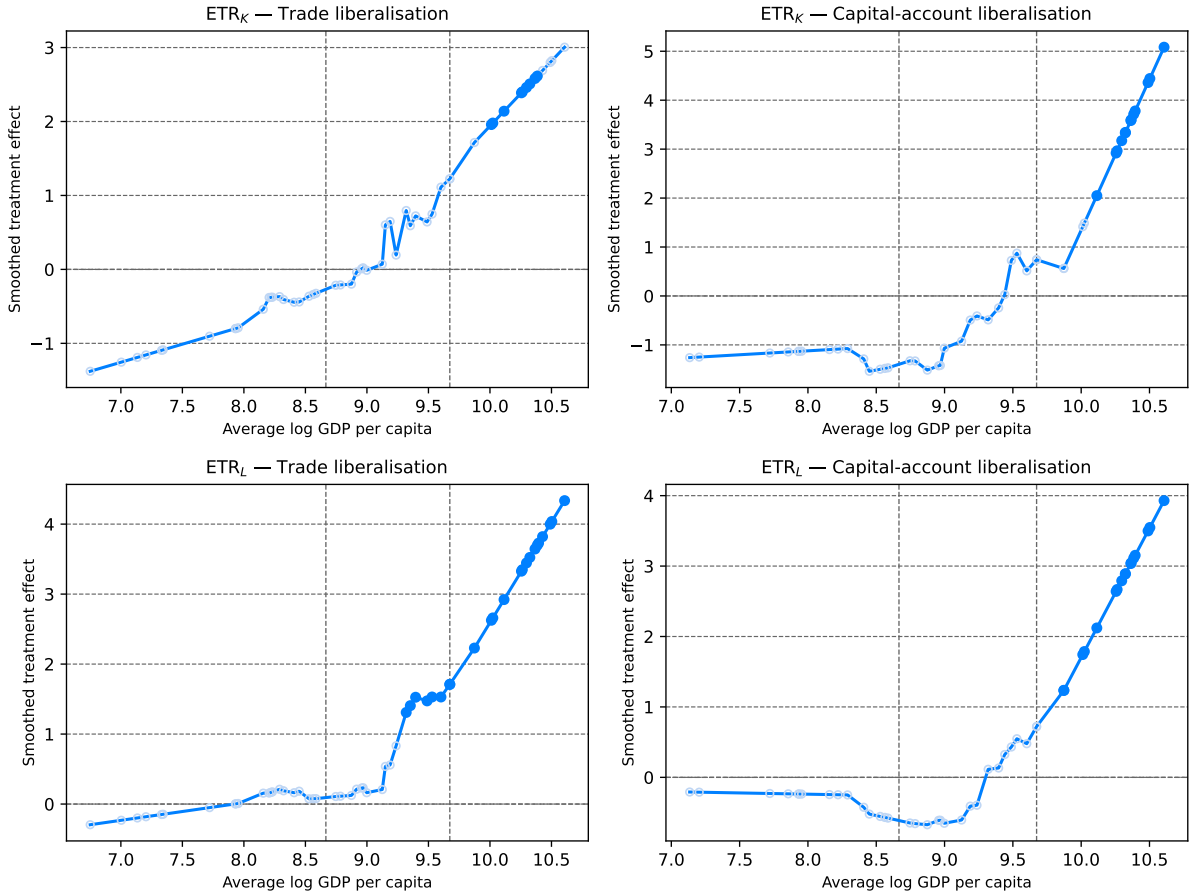


Figure D-1: The gradient of globalisation's effect on factor taxation, against income per capita. Constructed by the same method as Figure 3: each panel is a running-line smoother of the country-specific CCE treatment effect $\hat{\beta}_i$ on the country's average log GDP per capita, at a common span chosen by pooled cross-validation, for the indicated effective rate and form of liberalisation. Solid (hollow) markers denote points of the smoothed line (in)significantly different from zero at the 10% level; dashed vertical lines mark the two cut-offs separating the income terciles of the pooled treated distribution.

D.2 State capacity and income per capita

Capacity and income are related but are not the same ranking, which is what allows Table 6 to tell them apart at all. Figure D-2 plots the two country means against each other for the 142 countries with both. They correlate at 0.67 (Spearman, and 0.67 in levels), and the terciles they define agree for 63% of countries: a little over a third are placed differently by the two measures. Most of that disagreement is one step. Only four countries, 2.8% of the sample, sit in opposite terciles: three high-income economies fall in the bottom capacity tercile and one low-income economy reaches the top. The overlap is therefore substantial without being coincidence, and it is why the gradient traced against income (Figure D-1) resembles the capacity gradient of Figure 3 without reproducing it.

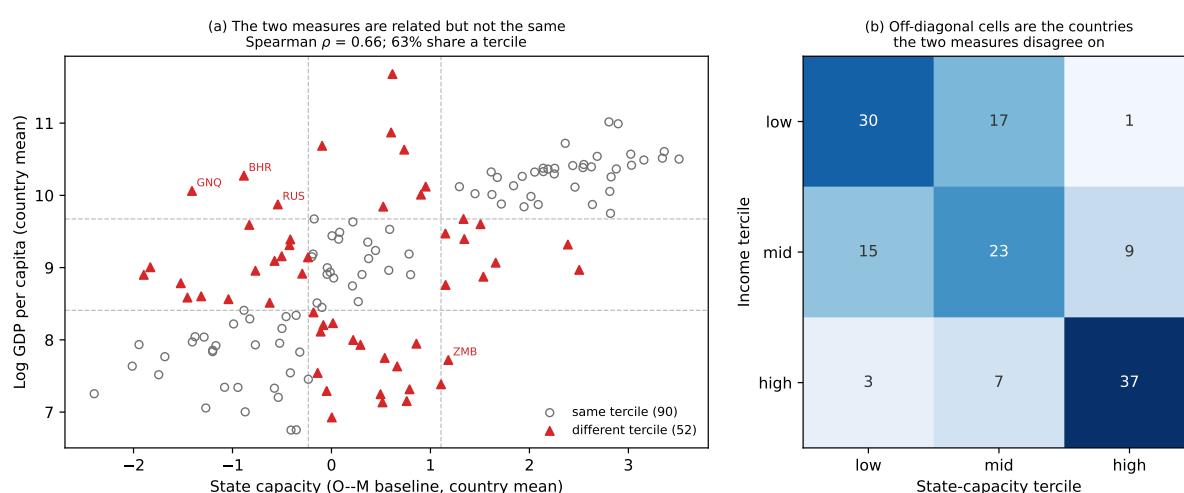


Figure D-2: State capacity against income per capita, country means. Panel (a) plots the O'Reilly and Murphy (2022) baseline state-capacity index against mean log GDP per capita; dashed lines mark the tercile cut-offs of each measure. Hollow circles are countries the two measures place in the same tercile, filled triangles those they place differently, with the four countries in opposite terciles labelled. Panel (b) gives the tercile cross-tabulation, so the off-diagonal cells count the countries on which the two measures disagree.

D.3 An alternative state-capacity index

Our headline moderator is the O'Reilly and Murphy (2022) baseline state-capacity index, which excludes any fiscal indicator. To check that the gradient is not an artefact of that particular construction, we re-estimate it on the latent state-capacity measure of Hanson and Sigman (2021), the posterior mean of a Bayesian latent-variable model that pools more than twenty indicators spanning the *extractive* (fiscal), *coercive*, and *administrative* dimensions of the state, over 1960–2015. Built from different inputs and a different method than the O'Reilly and Murphy (2022) baseline, which draws on V-Dem measures of impartial administration, the rule of law, and territorial reach, it nonetheless correlates with it at +0.87 across the 141 countries covered by both measures. Table D-1 re-runs the headline per-country CCE-DID

and the robust capacity gradient on each index in turn. The negative-to-positive pattern is unchanged: under both indices the predicted effect runs from negative or zero at low capacity to positive at high capacity, and the gradient is positive and significant in every specification. With both indices standardised, the per-standard-deviation gradients are directly comparable and similar in magnitude (for the capital effective rate under trade, 2.05 versus 1.74 per standard deviation), and the predicted low- and high-capacity effects, in points of the effective rate, are very close across the two. Because the Hanson–Sigman index includes an extractive/fiscal dimension that our baseline deliberately omits, a surviving gradient here is reassuring robustness rather than a tax-on-tax artefact of the moderator.

Table D-1: The capacity gradient under an alternative state-capacity index.

Capacity	Baseline (O–M)				Hanson–Sigman			
	Gradient $\hat{\gamma}_1$	low	med	high	Gradient $\hat{\gamma}_1$	low	med	high
Trade liberalisation								
ETR _K	+2.05*** (0.77)	−1.97** (0.82)	−0.06 (0.69)	+2.66* (1.42)	+1.74*** (0.65)	−1.66** (0.82)	−0.09 (0.67)	+2.25* (1.18)
ETR _L	+1.35*** (0.33)	−0.34 (0.41)	+0.93*** (0.34)	+2.71*** (0.61)	+1.33*** (0.37)	−0.24 (0.37)	+0.97*** (0.34)	+2.76*** (0.70)
Capital account liberalisation								
ETR _K	+1.93*** (0.68)	−2.05** (1.03)	−0.24 (0.76)	+2.32** (1.13)	+1.75** (0.70)	−1.82* (1.04)	−0.24 (0.77)	+2.12* (1.16)
ETR _L	+1.01*** (0.35)	−0.59 (0.46)	+0.35 (0.31)	+1.68*** (0.55)	+1.23*** (0.35)	−0.80* (0.44)	+0.32 (0.31)	+1.98*** (0.57)

Notes: Standard errors in parentheses. Each block re-runs the headline per-country CCE-DID and the Huber-M capacity gradient $\hat{\gamma}_1$ with the indicated state-capacity index as the moderator. Each index is standardised, so $\hat{\gamma}_1$ is the change in the effect per standard deviation of that index and the two blocks are directly comparable (as in Table 6); $\hat{\beta}$ is the predicted effect at the low, medium and high state capacity-tercile centroids (in points of the effective tax rate). Baseline (O–M) refers to the [O'Reilly and Murphy \(2022\)](#) fiscal-excluded index used throughout the main text. Hanson–Sigman: the latent state-capacity index of [Hanson and Sigman \(2021\)](#). The identification diagnostic is not repeated: the alternative index enters only the second stage, so the country residuals, and hence the Δ CSD test of Table 2, are unchanged. Statistical significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

D.4 Choice of cross-sectional averages

Our baseline specifications uses cross-sectional averages over the full sample for the conditioning variables ($\overline{lgdppc}$, $\overline{inv}$) and over the never-open control pool for the outcome ($\bar{y}^c$), the configuration the Δ CSD test of Section 4.4 selects. Table D-2 reports the capacity gradient under seven averaging schemes, together with the mean absolute Δ CSD across the four specifications. The capital gradient (ETR_K) is positive and significant under every scheme, for trade and for the capital account alike, though in two of the capital-account schemes only at the 10% level. The labour gradient is stable as long as the *outcome* average is taken over the control pool; it attenuates only when that average is formed over the full sample (the rows whose third entry is *F*), exactly the contamination the control-only

construction is designed to avoid: a full-sample outcome average absorbs part of the treated countries' own response. Every configuration is ΔCSD -clean ($|\Delta\text{CSD}|$ between 0.022 and 0.029), so the choice among them is not an identification question but a contamination one, and it bears mainly on the labour specifications.

Table D-2: The capacity gradient across cross-sectional-average configurations.

CA pool ($\overline{\text{lgdppc}}, \overline{\text{inv}}, \bar{y}^c$)	Capacity gradient $\hat{\gamma}_1$				$ \Delta\text{CSD} $
	ETR_K^{tr}	ETR_L^{tr}	ETR_K^{cap}	ETR_L^{cap}	
baseline (F,F,C)	+1.42*** (0.54)	+0.94*** (0.23)	+1.34** (0.47)	+0.72*** (0.24)	0.023
all full (F,F,F)	+1.64*** (0.50)	+0.43* (0.26)	+1.01* (0.44)	+0.31 (0.28)	0.022
all control (C,C,C)	+1.41*** (0.57)	+0.84*** (0.27)	+1.18* (0.59)	+0.70*** (0.29)	0.029
(F,C,C)	+1.40*** (0.55)	+0.85*** (0.25)	+1.13** (0.47)	+0.68*** (0.21)	0.025
(C,C,F)	+1.55*** (0.42)	+0.27 (0.23)	+0.87 (0.46)	+0.27 (0.33)	0.028
(F,C,F)	+1.59*** (0.49)	+0.31 (0.23)	+0.92 (0.46)	+0.23 (0.27)	0.023
(C,F,C)	+1.38** (0.53)	+0.89*** (0.28)	+1.29** (0.47)	+0.72*** (0.29)	0.025

Notes: Standard errors in parentheses. Each row is a configuration of the three cross-sectional averages ($\overline{\text{lgdppc}}, \overline{\text{inv}}, \bar{y}^c$), where F denotes an average over the full sample and C over the never-open control pool; “baseline (F,F,C)” is the headline. Entries are the Huber-M state capacity gradient $\hat{\gamma}_1$ for each cell; the final column is the mean absolute ΔCSD across the four cells. Statistical significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

D.5 Persistence of the state capacity ranking

We treat average state capacity as a slow-moving structural characteristic and use the country's full-period mean as the moderator. Figure D-3 shows this is warranted: capacity rankings are highly persistent. Comparing each liberaliser's 1980s and 2010s average capacity, the Spearman rank correlation between the two is 0.74 and 61% of the 67 liberalisers remain in the same capacity tercile; movement between the bottom and top terciles is rare, with only one country crossing in each direction. The year-by-year cross-country ranking of capacity stays correlated with the full-period ranking at around 0.9 throughout. Because the ranking that defines low- and high-capacity states barely reshuffles, the time-invariant moderator neither mechanically generates the gradient nor is sensitive to the window over which capacity is measured.

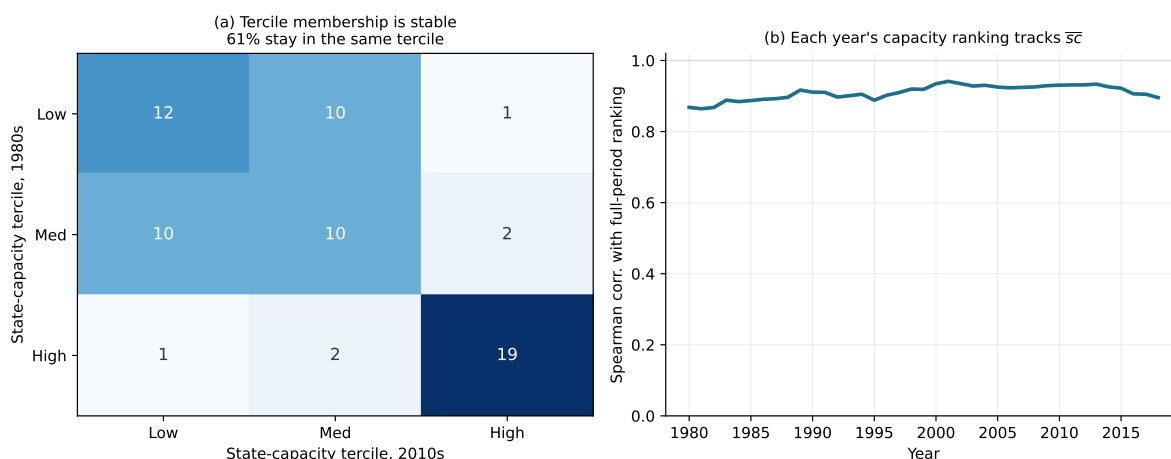


Figure D-3: Persistence of the state-capacity ranking among liberalisers. (a) Tercile transition matrix between each country's 1980s and 2010s average capacity; (b) the year-by-year cross-country rank correlation of capacity with the full-period mean.

D.6 Timing of the capacity measure

The predetermination check in Section 5.3 already rebutted reverse causation using each country's *pre-onset fiscal* capacity (the component most exposed, since opening most plausibly builds tax administration). We extend it here to the headline *baseline* index and to the full tercile pattern, since the full-period mean otherwise spans post-liberalisation years. Table D-3 re-estimates the gradient using two *pre-determined* timings of the baseline index: the country's mean capacity over the 1980s, before almost every liberalisation onset, given the post-1990 concentration of openings, and its capacity in the onset year itself. The negative-to-positive pattern survives both. The slopes stay positive and significant in every cell, attenuating only with the onset-year measure: the ETR_K trade gradient is +1.42 on the full-period mean, unchanged at +1.42 on 1980s capacity, and +1.04 on capacity at onset; the capital-account gradients move little (+1.34, +1.25 and +1.27). Capacity that pre-dates the opening cannot have been raised by it, so the gradient is not an artefact of reverse causality or of averaging over the post-treatment period.

D.7 Sample pruning

Table D-4 varies the two rules that determine which liberalisers enter the estimation. The top block changes the minimum-observations requirement from 15 to 30 on the baseline sample; the 25-observation row is the baseline. It reproduces Table 2 exactly. The bottom block instead *admits* the reverting and short-onset liberalisers the headline excludes, reconstructed from the raw openness indices on the same pooled-median bars. The capital and labour gradients are positive and significant under every rule. Tightening or loosening the observation bar moves the ETR_K trade gradient only within $[+1.37, +1.50]$, and admitting *all* reverters,

Table D-3: The capacity gradient under alternative timings of the moderator.

	(1)	(2)	(3)	(4)
State capacity	Gradient $\hat{\gamma}_1$	$\hat{\beta}(\text{low})$	$\hat{\beta}(\text{med})$	$\hat{\beta}(\text{high})$
<i>State capacity full-period mean (baseline)</i>				
ETR _K trade liberalisation	+1.42*** (0.54)	−1.97** (0.82)	−0.06 (0.69)	+2.66* (1.42)
ETR _L trade lib.	+0.94*** (0.23)	−0.34 (0.41)	+0.93*** (0.34)	+2.71*** (0.61)
ETR _K cap-acc liberalisation	+1.34*** (0.47)	−2.05** (1.03)	−0.24 (0.76)	+2.32** (1.13)
ETR _L cap-acc liberalisation	+0.70*** (0.24)	−0.59 (0.46)	+0.35 (0.31)	+1.68*** (0.55)
<i>State capacity in the 1980s</i>				
ETR _K trade liberalisation	+1.42*** (0.49)	−2.02** (0.89)	−0.09 (0.71)	+2.89** (1.33)
ETR _L trade liberalisation	+0.84*** (0.23)	−0.25 (0.41)	+0.90** (0.35)	+2.66*** (0.67)
ETR _K cap-acc liberalisation	+1.25*** (0.35)	−1.79** (0.88)	−0.09 (0.71)	+2.53** (1.00)
ETR _L cap-acc liberalisation	+0.56*** (0.20)	−0.39 (0.42)	+0.38 (0.31)	+1.56*** (0.50)
<i>State capacity in the onset year</i>				
ETR _K trade lib.	+1.04*** (0.38)	−1.77** (0.77)	+0.13 (0.67)	+2.18* (1.18)
ETR _L trade lib.	+0.69*** (0.21)	−0.23 (0.42)	+1.04*** (0.36)	+2.42*** (0.64)
ETR _K cap-acc lib.	+1.27*** (0.47)	−1.97* (1.07)	+0.39 (0.78)	+2.21** (1.09)
ETR _L cap-acc lib.	+0.60** (0.26)	−0.46 (0.54)	+0.66** (0.31)	+1.52*** (0.52)

Notes: Each panel re-runs the baseline country CCE and the Huber-M capacity gradient $\hat{\gamma}_1$ with the baseline state-capacity index measured at a different time: the full-period country mean (baseline), the country's 1980s mean, and its value in the liberalisation onset year (liberalisation-specific for the trade and capital cells). $\hat{\beta}(\text{low})$, $\hat{\beta}(\text{med})$ and $\hat{\beta}(\text{high})$ are the predicted effects at the low, medium and high capacity-tercile centroids, each with a significance star from a delta-method standard error. Statistical significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

which raises the trade sample from 55 to 60 countries, leaves it at +1.35. The pruning rules govern which countries enter, not the gradient they trace.

Table D-4: The capacity gradient under alternative sample-pruning rules.

Sample-pruning rule	Capacity gradient $\hat{\gamma}_1$			
	ETR_K^{tr}	ETR_L^{tr}	ETR_K^{cap}	ETR_L^{cap}
baseline (min 25, permanent)	+1.42*** (0.54)	+0.94*** (0.23)	+1.34** (0.47)	+0.72*** (0.24)
min obs ≥ 15	+1.37*** (0.50)	+0.99*** (0.23)	+1.09** (0.38)	+0.62*** (0.22)
min obs ≥ 20	+1.40*** (0.49)	+1.01*** (0.23)	+1.12** (0.41)	+0.73*** (0.21)
min obs ≥ 30	+1.51*** (0.55)	+0.98*** (0.23)	+1.43*** (0.46)	+0.82*** (0.25)
+ reverters, ≤ 1 reversal	+1.37*** (0.53)	+0.98*** (0.24)	+1.28*** (0.46)	+0.72*** (0.23)
+ reverters, any	+1.34*** (0.54)	+0.97*** (0.24)	+1.28*** (0.46)	+0.72*** (0.23)

Notes: Standard errors in parentheses. Huber-M capacity gradient $\hat{\gamma}_1$ for each cell. Top block: the headline treated sample at minimum-observation thresholds from 15 to 30 (the 25 row is the headline). Bottom block: the headline sample augmented with the ever-open countries the permanence rule excludes (onsets reconstructed from the pooled-median openness dummies), admitting those with at most one open→closed reversal and then any number. Tercile centroids are held fixed across rows. Stars: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

D.8 Unconditional trends with liberalisation-specific controls

Figure D-4 complements Figure 2 by comparing each group of liberalisers with its *own* never-open control (trade liberalisers against never-open-on-trade countries, capital-account liberalisers against never-open-on-capital countries) and re-basing every series to 1990 = 100. After 1990, the liberaliser and control series move together; if anything the controls rise by more (by 2018 the capital effective rate of the never-open controls stands at 125–127 against 110–121 for liberalisers, and the labour-rate gap is wider still). The unconditional rise in effective taxation is thus broadly shared and is not, on average, what opening up did. This is why its effect must be isolated by a conditional, capacity-graded design rather than read off the raw trends.

D.9 The 1990s watershed

Bachas et al. (2023) treat the 1990s as a watershed, while our panel begins in 1970. A reader may therefore ask whether the capacity gradient is a feature of the earlier decades rather than of the liberalisation era. It is not. The median liberalisation we estimate on occurs in 1991 for trade and 1993 for the capital account, and most of the identifying variation is already

Effective tax rates re-based to 1990, liberalisers vs margin-specific controls

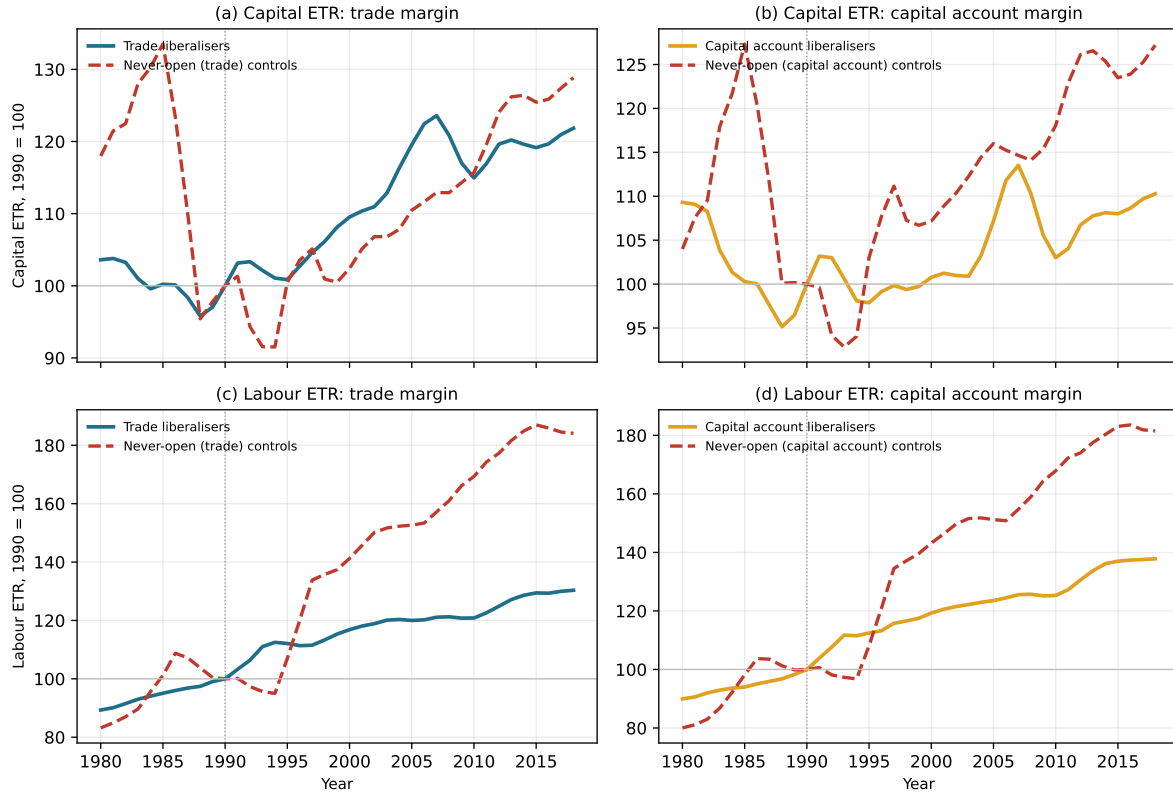


Figure D-4: Effective tax rates of liberalisers versus liberalisation-specific never-open controls, each re-based to 1990 = 100. Rows: capital and labour effective rates; columns: trade and capital-account liberalisation.

post-watershed: 35 of the 55 trade reforms and 38 of the 50 capital-account reforms fall in 1990 or later — see Table D-5 for details. Table D-6 presents results from a second-stage regression where we allow the capacity-gradient to differ by onset timing (before 1990, 1990–2018). All gradients for the post-1990 period emphasised by Bachas et al. (2023) are large and statistically significant, the pre-1990 ones less consistently so (ETR_L large and significant, ETR_K smaller and insignificant), in part due to the smaller sample sizes. Importantly, the difference between the two is statistically insignificant for all four specifications, indicating that our main findings are *not* driven by pre-1990 liberalisations.

Table D-5: Distribution of liberalisation onsets

	≤ 1984	1985–89	1990–94	1995–99	≥ 2000	median
Trade	11	9	18	9	8	1991
Capital account	9	3	14	12	12	1993

Notes: Number of liberalisation onsets falling in each period, over the estimation sample (the 55 trade and 50 capital-account liberalisers of Table D-6). The final column gives the median onset year. Most reforms fall in or after 1990: 35 of the 55 trade onsets and 38 of the 50 capital-account onsets.

Table D-6: The capacity gradient by timing of liberalisation

	Liberalised before 1990		Liberalised 1990–2018		Difference
	Gradient $\hat{\gamma}_1$	N_{tr}	Gradient $\hat{\gamma}_1$	N_{tr}	(pre – post)
Trade liberalisation					
ETR _K	+1.15 (0.61)	20	+1.45** (0.97)	35	-0.31 (1.14)
ETR _L	+0.96*** (0.26)	20	+0.70*** (0.33)	35	+0.26 (0.42)
Capital account liberalisation					
ETR _K	+1.02 (0.58)	12	+1.41** (0.70)	38	-0.39 (0.91)
ETR _L	+1.11** (0.21)	12	+0.45** (0.52)	38	+0.66 (0.56)

Notes: The capacity gradient $\hat{\gamma}_1$ (the robust slope of the per-country CCE treatment effect on average state capacity) for liberalisers whose reform began in 1990 or later and for those whose reform began before 1990. Both gradients and the difference between them come from a single robust regression with a pre-1990 $\times$ capacity interaction; the ‘Difference’ column is that interaction coefficient (pre-1990 minus 1990–2018), whose significance is the test of equal gradients. The first stage is unchanged from Table 2. Standard errors in parentheses; N_{tr} is the number of liberalising countries in each group. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

E Cross-sectional dependence diagnostics

Section 4.4 distinguishes cross-sectional dependence that is benign for identification, a common factor loading equally on treated and control units, from dependence that is treatment-correlated and therefore biasing. The Δ CSD test targets the latter; the Pesaran (2004) CD test and its bias-corrected variant CD* (Pesaran and Xie 2023) target the former, rejecting on any dependence at all. This appendix verifies, by Monte Carlo simulations calibrated to the panel’s own dimensions, that each statistic behaves as we claim.

We draw residual panels

$$r_{it} = g_i f_t + \delta h_t \mathbf{1}[i \text{ treated}] + e_{it},$$

at the trade-sample dimensions ($N_t = 55$, $N_c = 31$, $T = 40$, a quarter of observations missing at random). The common factor f_t loads on every unit; the treated-only factor h_t , present when $\delta > 0$, makes the dependence treatment-correlated. Symmetric-factor rows (i.e. symmetric factors between treated and control samples) set $\delta = 0$ and vary the common loading; treatment-correlated rows add $\delta = 0.5$ (mild) and $\delta = 1.0$ (strong), lifting the treated group’s mean absolute residual correlation about 0.08 and 0.24 above the control group’s correlations. Each row of Table E-1 averages 300 replications, with Δ CSD p -values from 149 label permutations under a fixed seed.

Table E-1: Size and power of the cross-sectional-dependence diagnostics.

Scenario	(1) Mean $ \rho $	(2)	(3)	(4) ΔCSD reject %	(5) reject %	(6) Mean $\hat{\beta}$
	control	treated	CD rej. %	5%	10%	(true = 0)
<i>Panel A: unspanned factor injected directly into the residual</i>						
No common factor (well specified)	0.18	0.18	4.7	4.7	8.3	
Symmetric factor, strong (TWFE-like)	0.41	0.41	100.0	5.3	10.0	
Symmetric factor, weak (CCE-like)	0.25	0.25	100.0	5.3	8.3	
Treatment-correlated, mild ($\delta = 0.5$)	0.32	0.40	100.0	55.3	65.0	
Treatment-correlated, strong ($\delta = 1$)	0.33	0.56	100.0	99.3	99.7	
<i>Panel B: unspanned factor enters through the outcome; first stage includes D_{it}</i>						
No factor (size)				7.0	13.0	−0.00
Injected into residual (as Panel A)				100.0	100.0	−0.00
Aligned with onset timing				17.0	29.0	+0.90
Misaligned with onsets				86.0	93.0	+0.73
Aligned with onset timing, strong ($\delta = 2$)				89.0	94.0	+1.80

Notes: Monte Carlo on simulated residual panels $r_{it} = g_i f_t + \delta h_t \mathbf{1}[i \text{ treated}] + e_{it}$ at the trade-sample dimensions ($N_t = 55$, $N_c = 31$, $T = 40$, a quarter of cells missing at random), 300 replications per row, fixed seed. ‘Mean $|\rho|$ ’ is the mean absolute pairwise residual correlation within the control and treated groups. ‘CD rej. %’ is the rejection rate of the [Pesaran \(2004\)](#) CD test at the 5% level. ‘ ΔCSD reject %’ is the rejection rate of the permutation test of equation (11) at the 5% and 10% levels (149 permutations). In Panel A, the no-factor row sets $\mu_g = \sigma_g = \delta = 0$; the symmetric-factor rows raise the common loading with $\delta = 0$ (benign dependence); the treatment-correlated rows add $\delta > 0$ (biasing dependence). Under a symmetric factor ΔCSD holds its size while CD always rejects; under a treatment-correlated factor ΔCSD gains power. The bias-corrected CD*, reported in Table 4, rejects 9.7% of clean-residual replications against the 5% nominal level and retains full power when dependence remains. Panel B routes the same factor through the *outcome* instead and runs the actual first stage (constant, D_{it} , control cross-sectional average) before computing ΔCSD ; it also reports the mean per-country effect $\hat{\beta}$ against a true effect of zero (400 replications, 199 permutations). Because that first stage contains D_{it} , a factor aligned with onset timing is absorbed into $\hat{\beta}$ and leaves the residual looking clean: the test then rejects only 17% of the time while $\hat{\beta}$ is biased by +0.90. Power returns when the factor is injected into the residual instead (100%) or is misaligned with the onsets (86%), and the test holds its size (7%) under the null.

The test holds its size against benign dependence and gains power against biasing dependence, see columns (4) and (5). Under a symmetric factor it keeps its nominal size however strong the factor: when the factor lifts the mean absolute residual correlation to 0.41, close to the 0.43 two-way fixed effects leave in Table 4, ΔCSD still rejects only 5.3% of the time against a 5% level. The high ΔCSD p -values of the mean-group and two-way-fixed-effects estimators in Table 4 are therefore the test working, not failing. Once the factor is treatment-correlated the test gains power, rejecting 55% of the time under a mild component and 99% under a strong one. The pattern survives at balanced-panel and small-control ($N_c = 15$) dimensions: size 2 to 8%, power 46 to 100% (not shown).

The experiments so far add the treated-only factor directly to the residual, the right test for a factor the estimator never touches, but not the only route an unspanned factor can take. If an unspanned factor belongs to the *outcome* and its timing aligns with liberalisation, the first-stage regression that forms each per-country residual, which includes the treatment dummy D_{it} , absorbs it into $\hat{\beta}_{f,i}$ and leaves the residual clean. Routing the same factor

through the outcome and running the actual first stage gives exactly that failure (Panel B): ΔCSD rejects only 17% of the time, little more than its 5% size, so it almost never flags the contamination, while the per-country effect it should have flagged carries a spurious $+0.9$ on a true effect of zero. The test is at its weakest precisely where the bias is largest. Against the residual-injected factor of Table E-1, or one whose timing is *misaligned* with the onsets, power returns (86 to 100%) and size holds (7%). We think this failure is unlikely to arise here, because the onsets are staggered. A single global shock moves at one date, so it cannot line up with more than a few countries' openings; for the rest it is misaligned, and that is the case the test does catch. To evade the test across the board, a confounder would have to be timed to each country's *own* opening, which means a separate shock per country rather than a common one. ΔCSD is therefore a necessary check and not a sufficient one, and we do not rest the case on it. We read it instead as one of several independent pieces of evidence that point the same way: the five-estimator horse-race (Table 3), the random-capacity placebo, and the direct evidence in Appendix C.3 that the leading candidate confounder, a coincident capacity-graded value-added-tax reform, is absent.

Table E-2: ΔCSD and capacity gradient bias.

Contamination	(1) True γ_1	(2) Mean $\hat{\gamma}_1$	(3) Bias	(4) ΔCSD reject %
None (factor spanned by controls)	1.00	1.00	+0.00	8
Treatment-correlated, mild ($\delta = 0.5$)	1.00	1.53	+0.53	100
Treatment-correlated, strong ($\delta = 1$)	1.00	1.99	+0.99	100

Notes: End-to-end DID + CCE Monte Carlo with a known true gradient $\gamma_1 = 1$, at the trade-sample dimensions ($N_t = 55$, $N_c = 31$, $T = 40$), 300 replications per row, fixed seed. Each treated country has effect $\theta_i = \gamma_1 c_i$; all units load on a common factor spanned by the control cross-sectional average. 'None' adds no further structure. The treatment-correlated rows add a treated-only factor with loading equal to capacity c_i and a path shifting at the treatment date ($\mu = 0.5$ mild, $\mu = 1.0$ strong), which the control average cannot span. 'Mean $\hat{\gamma}_1$ ' is the average Huber-M second-stage gradient; 'Bias' is its deviation from the truth; ' ΔCSD reject %' is the rejection rate of the permutation test of equation (11) at the 5% level. In this design ΔCSD holds its size when the gradient is unbiased and rejects in every replication when it is biased.

For the *level of dependence* we report CD^* rather than the raw CD (Table 4, Panel C). The CD column here shows why: it rejects in 4.7% of replications when no factor is present, but in 100% once any factor is, benign or treatment-correlated, strong or weak. It cannot separate the dependence that threatens identification from the dependence that does not, the over-rejection documented by Pesaran and Xie (2023), whose bias-corrected CD^* is calibrated on independent residuals while retaining power where dependence remains. That is the statistic we use for the level.

A final experiment confirms that the dependence ΔCSD detects is the dependence that biases the estimate. We simulate the full two-stage estimator with a known gradient $\gamma_1 = 1$: each treated country has effect $\theta_i = \gamma_1 c_i$, and all units load on a common factor the control cross-sectional average spans. The contaminated rows add a treated-only factor loaded by

capacity and correlated with onset timing, which the control average cannot remove. Results are presented in Table E-2. When the factor is spanned, $\hat{\gamma}_1$ is unbiased and ΔCSD rejects 8% of the time; when it is not, $\hat{\gamma}_1$ is biased upward by 0.53 (mild) to 0.99 (strong) and ΔCSD rejects in every replication. In this design the test rejects when, and only when, the treatment-correlated factor biases the gradient; Panel B above is the case it misses.

F A timing placebo for the capacity gradient

High-capacity liberalisers enter the sample on steeper tax trajectories than low-capacity ones, so the capacity gradient could in principle reflect a pre-existing, capacity-graded trend rather than liberalisation itself. We test this with a date-randomisation placebo. For each treated country we set aside its true onset and draw a fake one uniformly from its own observed span (retaining at least eight pre-onset and five post-onset years), rebuild the step dummy, and re-estimate the entire design: the per-country control-CA CCE-DID and the robust second-stage gradient. Repeating this 999 times under a fixed seed gives the distribution of the gradient's robust t -statistic under randomly placed onsets, against which we rank the true-onset value (one-sided p).

A pure state capacity-graded trend does not favour the true date. A step dummy placed anywhere on a linear trend carries the same coefficient, so under such a trend the gradient at a random date matches the one at the true date and the true onset is not special; only an effect concentrated at the opening makes it so. Figure F-1 shows the placebo as four null distributions, one per outcome and reform. In each panel the grey bars are the gradients from the 999 random-date placements and the vertical line is the true-onset gradient of Table 2, annotated with the one-sided placebo p : the share of random dates whose gradient matched or exceeded the true one. A p of 0.036 means that 35 of the 999 placements beat the true date; a p of 0.001 means that none did. In all four the true-onset gradient sits in the upper tail and clears the 5% level: the coefficient placebo p ranges from 0.001 to 0.036, and the equivalent test on the robust t -statistic from 0.001 to 0.046. The gradient is thus concentrated at the true openings rather than spread smoothly across dates, most clearly so on the labour rate.

Two features make the test informative here. Because the onsets are staggered, a single common factor cannot line up with every country's date, so a common trend cannot pass as a concentrated effect and is caught. And the placebo does not net out a confounder timed to each country's own opening; that residual case is the coincident country-specific reform, which Appendix C.3 closes with the consumption-tax evidence.

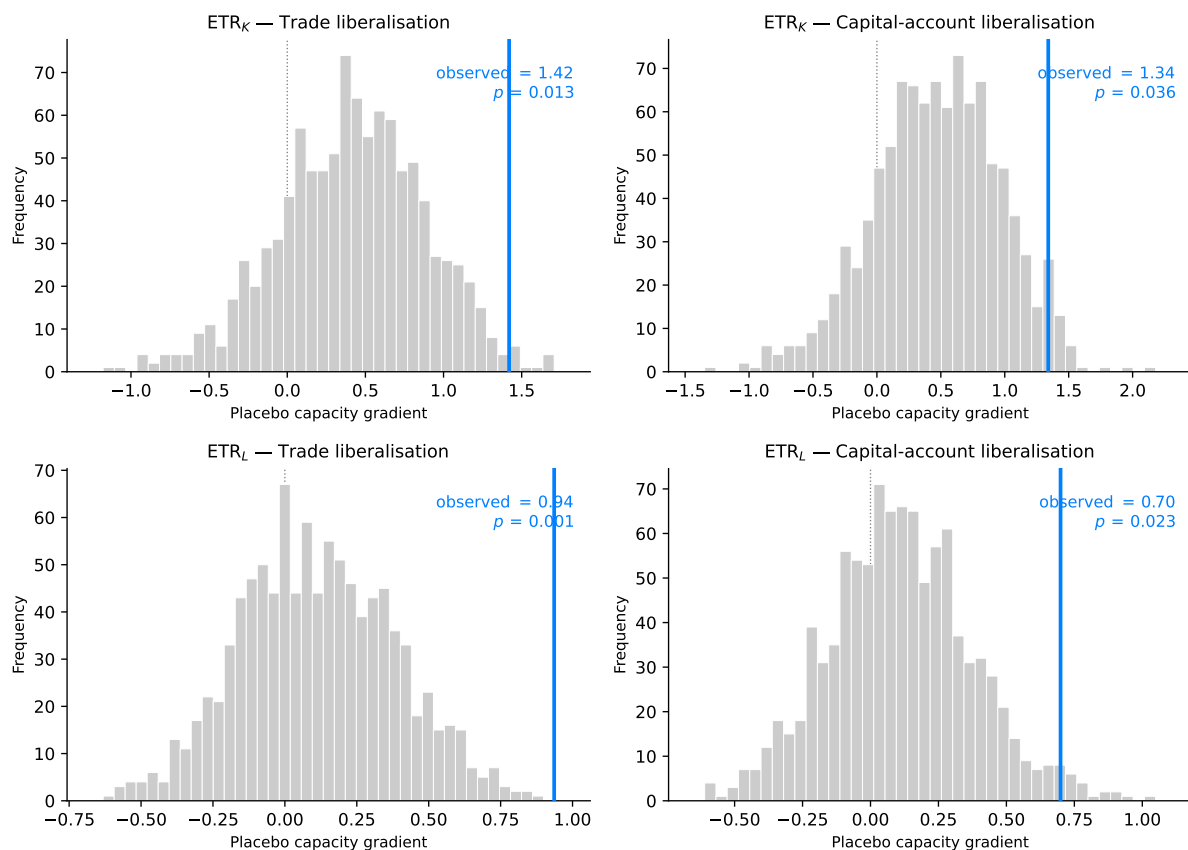


Figure F-1: Timing placebo: distribution of the capacity gradient under 999 random onset dates, by outcome and reform. Grey bars are the placebo gradients; the vertical line is the true-onset gradient of Table 2, annotated with the one-sided placebo p (the share of random dates matching or exceeding it). The placebo distribution is not centred at zero, because a randomly placed onset absorbs part of each country's secular trend; the true-onset gradient nonetheless falls in the upper tail in every panel.